

Accommodation, acquisition, attrition? The (un)changing grammars of Standard Southern British English speakers in Belfast*

Abstract

There has been traditionally little communication and consensus between the field of language attrition, and variationist sociolinguists investigating dialect change, despite shared interest in the outcomes of language and dialect contact. In this paper, we aim to bridge this divide, outlining ways in which the fields already converge, and where they can inform each other to build a greater understanding of the mechanisms underlying changes to an adult's otherwise established grammatical state. We do so by investigating two properties of Belfast English in the grammars of L1a Standard Southern British English speakers who have been living in Belfast for, on average, 25 years, having migrated from England after childhood. We find no use or acceptance of the Northern Subject Rule, but individual-level use and acceptance of embedded inverted questions. The study is framed by Hicks & Domínguez's (2020) Attrition via Acquisition model. We outline how this model of grammatical attrition can be used to predict where and how morphosyntactic change as a result of dialect (or language) contact may occur, and show how greater specification of this model with relevant factors from sociolinguistics can account for the results of the study at hand, concluding there is more to be gained from these fields working together than remaining disparate.

1 Introduction

Native language (L1) grammatical attrition is a process of change to the representation, processing or use of an L1 by adults who have migrated and settled in an environment in which a different language is the community language. As a result of processes including acquisition, increased exposure to, and use of, a second language (L2), and restriction of contact with the L1, it is possible for properties of a speaker's L1 grammar to become 'vulnerable' to change (e.g., Köpke & Schmid 2004, Schmid & Köpke 2017, Schmid & Dusseldorp 2010, Domínguez 2013, Gürel & Yilmaz 2011, Iverson 2012, Tsimpli et al. 2004). Although earlier definitions of language attrition focused on the 'loss' (of competence), 'disintegration' or 'unravelling' (Seliger & Vago 1991:3) of an L1, attrition is now more generally taken to refer to any change or modification to the language as a result of contact: 'an unconscious rearrangement or restructuring... but not as a drastic loss/decay' (Gürel & Yilmaz

* This work was funded by the UKRI (AHRC award ref. AH/...). Further acknowledgments to be added.

2011:222). Thus, in this paper, we refer to attrition as any modification of the adult grammar (see also Hicks & Domínguez 2020).

Work in language attrition has also shown evidence of morphosyntactic changes among speakers who migrate and subsequently receive extensive input from a second *dialect* as opposed to a second *language*, particularly in the case of Spanish-speaking communities in the United States, where speakers of Latin American and Caribbean varieties of Spanish frequently come into contact within a community (e.g., Domínguez 2013, Otheguy & Zentella 2012). Morphosyntactic change to mature adult grammars has also been documented in variationist sociolinguistics, generally under the term *long-term accommodation* (e.g., Trudgill 1986), though the focus of these variationist studies has tended to be on lexical or phonetic/phonological changes. However, there has been little discussion between variationist sociolinguistics and language attrition. The benefits of considering the two fields together have been recognised by some, such as Chambers (1992) in his paper on second dialect acquisition of Canadian teenagers in England, who states ‘if the principles of dialect acquisition gain empirical depth, their relationship to principles of first-language acquisition and second-language acquisition could open up a broad area of interaction among language researchers who have hitherto seldom recognized their common ground.’ (1992:703). However, more often, clear dividing lines have been drawn, such as by Trudgill (1986) and Kerswill & Williams (2002a:82): ‘far from being a limited “special case” of language contact, dialect contact is a phenomenon typologically different from language contact’.

We feel that this debate can benefit from a more rigorous approach to the ways in which research on grammatical attrition, and research on morphosyntactic change in the variationist framework, can inform each other to build a greater understanding of the mechanisms underlying changes to an adult’s otherwise established grammar. In this paper, we do so by testing a particular model of grammatical attrition (the *Attrition via Acquisition* model (AvA) (Hicks & Domínguez 2020)) in the investigation of potential changes to the grammars of speakers of L1a Standard Southern British English (SSBE) as a result of extensive contact with L1b Belfast English. We show how the model can make formal predictions about where, and why, changes to a speaker’s adult grammar may arise. Based on the results of the study of the SSBE speakers in Belfast, we subsequently show how variationist theories can further specify the AvA model, arguing that bringing together these two perspectives allows for a greater understanding of the nature of change to an adult individual’s grammar during their lifespan.

2 Background on attrition

2.1 Grammatical attrition in the bidialectal context

While L1 attrition is widely documented for various aspects of language—such as processing linguistic input (e.g. Dussias 2004), speech fluency (e.g. Bergmann et al. 2015), phonetic features

(e.g. de Leeuw et al. 2013, Ulbrich & Ordin 2014), overall accent perception (e.g. de Leeuw et al. 2010, Hopp & Schmid 2013), and lexical change (e.g. Yilmaz & Schmid 2012) — *grammatical* attrition, the modification of the representation of a morphosyntactic property within an individual's native grammar as a result of extensive exposure to and use of an L2, is a much more rarely attested phenomenon (e.g. Domínguez 2013, Gürel & Yilmaz 2011, Iverson 2012, Tsimpli et al. 2004). The extent to which L1 grammatical attrition takes place appears to be modulated by a range of individual and social factors. One of the most important factors is the age of acquisition of the L2 (Byland 2009, Köpke & Schmid 2004), with almost entirely wholesale grammatical change possible for children before the age of around 12, and considerably more limited effects found in mature, adult speakers who have acquired end-state L1 grammars. It is adult speakers like these that the study here focuses on. Beyond age of L2 acquisition, other social factors have been argued to have influence, such as length of time spent in the L2 community, education, type and level of exposure to the L2, and affective factors influencing alignment towards/away from the L1 or L2 community (e.g. Schmid & Dusseldorp 2010, Domínguez 2013, Köpke 2007, Schmid & Köpke 2017). The extent and possibility of grammatical attrition may also be modulated by linguistic factors—such as shared morphosyntactic properties (e.g. Tsimpli et al. 2004) and the degree of crosslinguistic similarity between the L1 and L2 (e.g. Schmitt 2019), with speakers of typologically closely related languages more likely to undergo attrition on morphosyntactic properties¹. Hicks & Domínguez (2020) argue that, taken to its natural conclusion, this suggest speakers of a dialect of an L1 in extensive contact with a second dialect of the L1 would be most susceptible to grammatical attrition, given that the grammatical features in question will be shared to a significant extent between the two varieties, but may also exhibit small differences.

Indeed, grammatical attrition as a result of input from a second dialect is attested in the literature (e.g. Cabrera-Puche 2008, Domínguez 2013, Otheguy & Zentella, 2012). Notably, these attested changes did *not* arise due to input from an L2. For example, Domínguez (2013) and Domínguez & Hicks (2016) show that L1 Cuban speakers of Spanish in Miami increased their proportion of use of null subjects as a result of contact with speakers of other dialects of Spanish, while L1 Spanish speakers in the UK—which does not have a significant community presence of Spanish resulting in substantial cross-dialectal contact (Cazzoli-Goeta et al. 2010)—did not see any changes to their use of null subjects. Similarly, Cuza (2010) found attrition to the present tense aspect system of L1 Caribbean Spanish speakers in the US, who are in regular community contact with speakers of other varieties of Spanish (e.g. Mexican Spanish). This result was not replicated for L1 Spanish speakers in the UK by Hicks et al. (2024), who showed no group level attrition for the same property, again in a context which lacks extensive community contact with other L1 varieties. While

¹ There is a paucity of research into the possibility of attrition between typologically dissimilar languages as compared to typologically similar languages; nevertheless, evidence from first and second language acquisition supports the idea that typological closeness can have an effect (e.g. Schwartz & Sprouse 1994, White 1989, Borragan et al. 2021).

these cases involved changes in the proportions of use of an existing feature in the speakers' grammars, there is also evidence of speakers acquiring a new feature or a new distribution of a feature. For example, Otheguy & Zentella (2012) find Cuban Spanish speakers in New York acquire a new distribution of overt and null subjects, based on extensive input from mainland Spanish varieties.

In all of these cases, input from a new dialect resulting in grammatical attrition has emerged as a result of emigration and the establishment of a new community of minority language L1 speakers within a majority L2 environment. However, migration to a community with a different dialect within an L1 context can also lead to morphosyntactic change. In the sociolinguistics literature, this is primarily understood as *accommodation* (Giles 1979)—an instance of linguistic convergence between a speaker and interlocutor(s) of a different variety in an individual speech situation. Less frequently discussed is the idea of *long-term accommodation* (Auer & Hinskens 2005, Trudgill 1986). Long-term accommodation occurs as a result of frequent short-term accommodation, leading the speaker to employ the 'converged' feature outside of an accommodating interactional context (Trudgill 1986:40)—in other words, this feature is now a part of the speaker's lexicon, phonetic system or grammar. This definition suggests a situation in which the speaker did not previously use the feature, but has acquired it through increased input and accommodating over time, and it is this we will focus on in the rest of the paper².

A number of works have investigated long-term accommodation in adults' phonetic/phonological systems (e.g. Trudgill 1986, Auer, Barden, and Grosskopf 1998, Conn & Horesh 2002, Evans 2004, Nycz 2016; see Siegel 2010 for an overview). In terms of morphosyntactic long-term accommodation, however, there is considerably less data, with a crucial exception in Kerswill (1994). In Kerswill's (1994) study, he examines the speech of rural dialect speakers from the Strilelandet area, the rural area surrounding Bergen in Norway. These speakers ('Strils') have moved to the urban centre of Bergen, and are experiencing extensive contact with the Bergen dialect. The relationship between the Strilelandet dialects and the urban Bergen dialect is unusual for two neighbouring areas in having a particularly strong bundle of isoglosses distinguishing the two varieties rather than operating on a dialect continuum (Kerswill 1994:36). For example, Norwegian has two standard written varieties, Bokmål and Nynorsk. In Bergen, Bokmål is the written standard and resembles the Bergen dialect quite closely, while in Strilelandet, Nynorsk is the written standard, which the Stril dialects map closely to. Drilling down, there are distinct morpho-lexical features which are part of the Bergen dialect, but are not found in the Stril dialects (Kerswill 1994:34), nor used in a style-shifting context (i.e. as an optional 'standard variant') by Stril speakers who have remained in the Strilelandet area (Kerswill 1994:77). Kerswill finds that the Strils living in Bergen

² Rather than changes in the frequency of use of a feature in someone who may already be bidialectal.

utilise ‘a considerable amount of morpho-lexical mixing’ (1994:146). For example, in the Stril dialects, there are three ‘weak’ verb classes indicated by *-a*, *-ə* or a null suffix, while in the Bergen dialect, there is only one present tense verb suffix, *-ər*. Strils, to varying extents, used features like the Bergen present tense suffix in conversation with each other and in an interview with a community outsider, suggesting that this is not simply accommodation in interaction with Bergen speakers.³

Variationist sociolinguistics has long considered the outcomes for an individual who migrates to an area where a different dialect is spoken as entirely independent from changes due to increased input in an L2 (e.g. Kerswill & Williams 2002a:82). As receiving input from a new dialect doesn’t involve speakers learning a new language, it is argued items can be mixed ‘at will’ with ‘minimal loss of intelligibility’ and without being subject to the equivalence constraint (Poplack 1980:586). However, some of the key evidence of change for individual dialect speakers as a result of migration has arisen from study of speakers of two different *languages* (see Nordenstam’s 1979 (cf. Trudgill 1986) study of lexical and morphological changes in the speech of Swedish L1 speakers living in Norway and ‘accommodating’ to L2 Norwegian)⁴. Furthermore, when we compare the factors argued to affect whether ‘native language attrition’ (due to extensive input from an L2) and ‘long term accommodation’ (due to extensive input from a second dialect) take place, many similarities arise:

1. Both require extensive exposure to and input from a second variety, whether it be a language or dialect.
2. Typological proximity determines the possibility of vulnerability; while this may align to the difference between *language* vs *dialect* contact, it may not.
3. There appears to be a shared cline of vulnerability for change, with lexical and phonetic/phonological properties more likely to undergo change than morphosyntactic properties.
4. The transparency of equivalent features between the two varieties impacts the possibility of grammatical adaptation.
5. The outcome of language or dialect contact is likely to be very different if the speaker in question is an adult, as opposed to a child.
6. Social factors such as networks and attitudes play a role in determining the outcome of contact.

And, despite the potential influence of each of these factors:

³ Eight of Kerswill’s participants moved to Bergen aged 16 or below; they were more likely to have undergone changes to their morpho-lexical system, and four had become fully bidialectal/could code-switch clearly between a Stril dialect and a Bergen dialect. For the remaining 31 participants, five mirrored the non-mobile Stril controls. The rest of the participants had shifted towards more Bergen features.

⁴ The problematic nature of using Swedish and Norwegian as key evidence for ‘dialect contact’ is also acknowledged in a footnote by Kerswill (1994:138).

7. Whether or not change takes place for any given speaker is highly individual, subject to considerable between-speaker variation.

Given these similarities between the factors held to affect change in native language attrition and long-term accommodation, we propose that instances of bidialectal language contact leading to long-term accommodation can also be considered within generative frameworks applied to native language attrition. We see this as a natural extension of the ongoing development of models of stable inter- and intra- speaker morphosyntactic variation in dialects, in which existing features and processes of Minimalist syntax can be exploited in order to account for the distribution of variants in a community and in an individual's speech (e.g. Henry 2005, Adger 2006, Adger & Smith 2010, Nevins & Parrott 2010; see Adger et al. in press for a summary). At the same time, generative models of first and second language acquisition have used the core principles of Minimalist syntax and UG to account for the patterns seen in language learning (e.g. Schwartz & Sprouse 1996, White 2003, Putnam & Sánchez 2013, Lidz & Gagliardi 2015, Slabakova 2016, Rothman & Slabakova 2018) and, more recently, in language attrition. While the tools of Minimalist syntax and UG are the same in both frameworks, studies in generative acquisition and attrition also bring a focus on the role of *input* and *intake* for an individual speaker (e.g. Carroll 2001, Truscott & Sharwood Smith 2011, Biberauer 2019), concerned with questions such as: what type of linguistic data triggers acquisition, or attrition, of a feature? At what point does input gain a representation (i.e. become intake)? How do the principles of UG interface with incoming linguistic data from a new variety? These questions are relevant to understanding 'long-term accommodation' in the grammars of individual adult speakers who are exposed to a second dialect. They provide a way to formalise assumptions from variationist study about the roles of typological proximity, exposure, and social networks, which can be used alongside the principles of UG as already applied within sociosyntax to develop a clearer picture of the accommodation/acquisition/attrition process.

In this paper, we test a particular model of generative language attrition on dialect data: the Attrition via Acquisition (AvA) model proposed in Hicks & Domínguez (2020). The model, grounded in Minimalist theory, makes predictions about when morphosyntactic attrition is likely to occur for adult speakers, given appropriate levels of input and particular morphosyntactic feature combinations. In the next section, we detail the properties of the AvA model and outline how testable predictions can be made based on this model which are relevant for study of change to adult speakers' grammars.

2.2 The Attrition via Acquisition model

The AvA model asserts that the mechanism by which L1 grammar undergoes unconscious change in instances of L1 grammatical attrition is the same mechanism by which a grammar changes during language acquisition. The AvA model, then, is principally a model of language *acquisition*, and indeed

is based on the L1 acquisition model of Lidz & Gagliardi (2015), extended and modified to incorporate Ln acquisition contexts. The model is schematised in Figure 1.

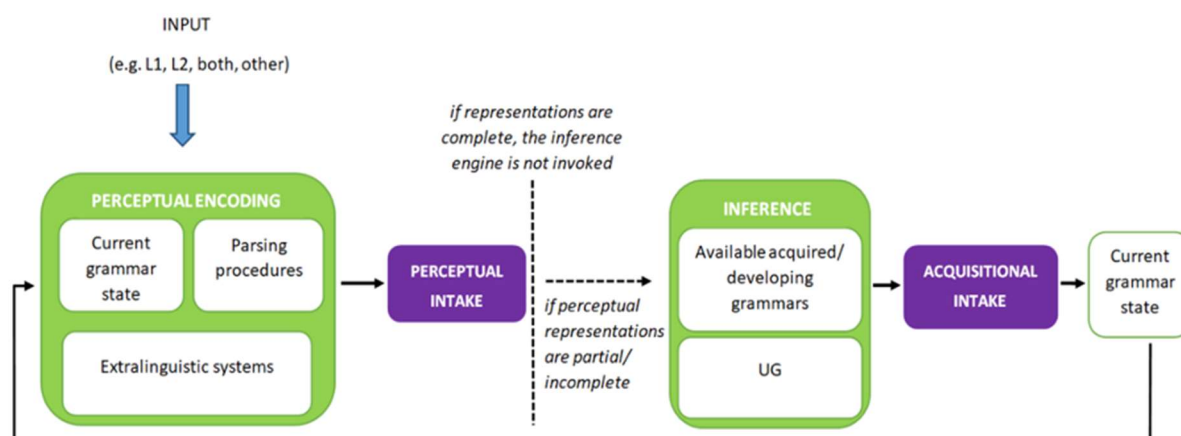


Figure 1: The AvA model as presented in Hicks & Domínguez (2020:152)

AvA incorporates both a grammatical mechanism by which representational change in the L1 grammar is instantiated, as well as an acquisitional mechanism that gives rise to those changes. The acquisitional mechanism is based on a distinction between linguistic input and ‘acquisitional intake’: an instruction to update the speaker’s existing grammar, a possible outcome of exposure to input brought about by a combination of grammar-internal and processing mechanisms. Following Lidz & Gagliardi, linguistic input is perceptually encoded based on a number of grammar-internal and grammar-external systems; a psycholinguistic process that leads to ‘perceptual intake’ of the utterance on various linguistic levels. When L1 linguistic input is received by a speaker with a stable, end-state grammar, perceptual intake representations of that input are complete; there is no acquisition. On the other hand, suppose that a speaker is exposed to linguistic input from a distinct language they are in the process of acquiring. Depending on the nature of that input and the state of the grammar that is used to process that input, the perceptual representation of the utterance may be incomplete or mismatching with the expectations based on the speaker’s current L2 grammar state. In this case, Hicks & Domínguez (2020) propose that the inference engine is invoked, and acquisitional intake is generated, leading the L2 grammar to update. L2-induced L1 attrition arises where the update to the L2 grammar as the result of acquisitional intake is also applied to a corresponding item of the L1 grammar. Next, suppose that the input that a speaker receives is not from a distinct L2, but from a dialect of the L1 that is substantively different from that of the speaker. The same mechanisms leading to acquisitional intake are invoked here, too, potentially resulting again in an adjustment to the speaker’s L1 grammar. Under the definition applied above, this also represents an example of L1 attrition, induced by extensive exposure not to input from a second language, but from a second dialect of the L1.

2.3 Why the AvA model?

Following Minimalist assumptions regarding the language faculty, this model allows detailed, linguistically informed predictions to be made regarding i) what grammatical properties which may be susceptible to attrition and ii) under exposure to what bilingual/bidialectal exposure conditions.

The AvA model allows L1 grammatical attrition to be understood as an unconscious representational change to a speaker's grammar arising through a combination of existing general mechanisms relevant to language acquisition, notably concerning input, perceptual processing thereof, and acquisitional inference. Hicks & Domínguez (2020) propose that a natural grammatical mechanism for this representational change is provided by a feature-based model of the syntactic component of the grammar, based on the Minimalist framework originally developed in Chomsky 2000, 2001 with variation in morphosyntactic output taken to be a result of variation in the availability of and syntactic properties of the features of the functional items in the lexicon (Borer 1984, Chomsky, 1995). Feature Reassembly (Lardiere 2008, 2009) is often applied to generative L2 acquisition in modelling the development of an individual's L2 grammar. The change to the L1 grammar that is induced by acquisitional intake thus entails a change to the feature specification of relevant functional items of the L1 grammar. Similarly, in the generative sociosyntactic literature the same framework has been used to analyse an array of inter- and intra-speaker variation in non-standard varieties (for an overview, see Adger et al., in press). So, the AvA model allows attrition to be understood as a natural outcome of exposure to a new type of input, and the mechanisms that instantiate it are compatible with existing generative approaches to morphosyntactic variation and change.

For example, Hicks & Domínguez predict that L2-induced attrition will arise for those grammatical properties that exhibit similar behaviour in the L1 and L2, but where fine-grained differences obtain between them. In a feature-based model of morphosyntax outlined above, this is encoded as variations in the specification of bundled morphosyntactic features on their corresponding items. A further prediction, advanced in Hicks et al. (2024), is that due to the mechanisms of attrition which update the L1 grammar by incorporating an acquired property of the L2 into the existing L1 grammar, attrition will more readily involve adding a grammatical option of the L2 rather than losing a grammatical option of the L1; see also a similar prediction of the 'Set-Theoretic Language Attrition Model' of Gürel (2002, 2007). Both predictions are consistent, for example, with the findings of Gürel & Yılmaz (2011), who study pronominal interpretations of Turkish L1 speakers following long periods of exposure to either Dutch or English as an L2 in the Netherlands and the USA. They find that the feature mapping of Turkish L1 speakers may change to permit a binding option not present in Turkish, but permitted in Dutch and English. With Turkish being a null subject language, the overt subject pronoun *o* is prevented from being locally bound by a subject (1b), while the equivalent interpretation is available for English (1a) and Dutch:

- (1) (a) Arzu_j believes that she_{j/k} is charming. (English)
- (b) Arzu_j [o-nun*_{j/k} cazibeli ol-duğ-u]-na inan-ıyor.
 Arzu s/he-_{GEN} charming be-_{NOM-3SGPOS-ACC} believe_{PROG}
 ‘Arzu_j believes that s/he_k is charming.’ (Turkish)
- (c) Arzu_j [o-nun_{j/k} cazibeli ol-duğ-u]-na inan-ıyor.
 ‘Arzu_j believes that s/he_{j/k} is charming.’ (attrited Turkish L1)

In experimental tasks, bilingual L1 Turkish speakers in the Netherlands and the USA selected the L1 option for *o* – an interpretation disjoint from the subject – less frequently than monolingual controls, a possibility added to their L1 grammar based on the corresponding property of the L2.

The AvA model also makes predictions about the nature of the input that may induce L1 attrition. For example, where full perceptual encoding of L2 input is available, grammatical properties of the input that bear out fine-grained differences from the speaker’s existing grammar are more readily ‘noticed’ and hence carried through to acquisitional intake. Hicks & Domínguez predict, then, that attrition will thus arise more readily where the L2 is a variety closely related to the L1, and less readily where the L1 and L2 are typologically distinct languages. For example, as detailed in section 2.1, there are instances of Spanish dialects speakers undergoing attrition in contact with other varieties of Spanish, while similar changes do not obtain when speakers are in contact with English, a more typologically distinct variety. Similar results are found by Schifano (2024) for Ecuadorian Spanish speakers in the UK in aspects of their mood and verbal systems. Furthermore, the use of evidence from L1 Swedish speakers exposed to L2 Norwegian input as evidence for ‘long-term accommodation’ in the variationist literature (see section 2.1) supports this idea of a cline of perceptibility and typological relatedness in the perceptual encoding module of the AvA model.

Having outlined the AvA model of grammatical attrition and how it provides a formal model for long-term grammatical accommodation that complements existing models of variation in sociosyntax, in section 3 we detail how we test the model’s predictions by looking at speakers of one dialect who have experienced long-term immersion in a second dialect.

3 Testing the predictions

To test the predictions of the AvA model detailed in section 2.2, we require two closely related varieties—namely, dialects of the same language—which nevertheless differ regarding some (morpho)syntactic properties. We investigate 30 speakers of L1a Standard Southern British English (SSBE) who have been living in Belfast for over 15 years, and thus receive regular and extensive input from speakers of Belfast English, an L1b. Key characteristics of both varieties are detailed in the

next sections, as well as the perceptual and attitudinal relationship between the two. Finally, we introduce the specific morphosyntactic properties investigated in this study.

3.1 Standard Southern British English

SSBE is the label now favoured by many linguists for something like the variety known as Received Pronunciation (RP) (e.g. Hughes, Trudgill & Watt 2012:3, Lindsey 2019:4). However, while traditional RP, as documented by e.g. Jones (1918), Gimson (1962) and Wells (1982), functioned as an idealised accent of (upper-class) ‘standard English’ to which non-standard accents and dialects could be compared, SSBE reflects a real, less socially-stratified variety, which has also undergone a number of diachronic changes (see Fabricus (2018) for an overview). Though there is some minor regional and social variation in the syntax of speakers in the south of England, this variation is ‘relatively homogenous’ across the region (Edwards 1993:216) and mainly includes ‘vernacular universals’ (Chambers 1995) such as non-temporal *never* (Cheshire et al. 1993, Kortmann & Szmercsanyi 2004) and negative concord (Anderwald 2005, Cheshire 1982). SSBE is thus generally assumed to be ‘standard English’ (Anderwald 2008:440).

Due to extensive dialect levelling in the south of England (Kerswill & Williams 2002b, Trudgill 1999), SSBE is spoken widely across the south and much of the Midlands. For the purposes of this study, we chose to include an L1a geographic cut-off point roughly following the FOOT-STRUT isogloss (Chambers & Trudgill 2004) in order to exclude speakers who may have been exposed to the relevant features in this study through input from northern British English varieties.

3.2 Belfast English

Belfast English is a variety of Northern Irish English, spoken in and around the capital. The English spoken in Belfast evolved from contact between English and Scottish settlers with the native Irish-speaking population during the plantations in the seventeenth century (Henry 1995:7)⁵, and in the present day is considered somewhat of an amalgamation of features of Ulster Scots (spoken in the north of Northern Ireland) and the Mid-Ulster dialects spoken across the rest of the region (Corrigan, 2010:18, Hickey 2007:333, Milroy & Milroy 1985:354). In terms of its syntax, Belfast English differs from standard English (or, SSBE) in a range of ways, investigated in detail by e.g., Henry (1995, 2005, 2016), Henry & Cottell (2007), Corrigan (2010). Many of these properties are shared with other varieties of English—including other Hiberno-English varieties, and Scots. This is true of both the Northern Subject Rule (NSR) and embedded inverted questions (EIQs), which are investigated in this study and detailed in sections 3.4.1 and 3.4.2.

⁵ Belfast no longer has a native Irish-speaking population, and has not since the nineteenth century. Influence from Irish on the variety is thus primarily historical, and not due to ongoing language contact.

3.3 Perceptions and identity between Belfast English and SSBE

The sociolinguistic relationship between Belfast English speakers and SSBE speakers is complex. For SSBE speakers in England, there is generally a level of distance to their perception of Northern Irish English varieties (e.g. Montgomery 2012, Wells 1982), meaning that speakers are unlikely to draw distinctions between, for example, a Belfast variety and a rural Northern Irish variety due to lack of exposure. To the extent that there is perception of a Northern Irish variety, it is considered highly attractive in popular opinion polls (Lucek & Garnett 2019), but it is not seen as prestigious, and SSBE speakers do not consider Northern Irish English as socially attractive as speakers of other ‘non-standard’ varieties of English in Great Britain do (e.g. speakers of Scottish or Welsh English) (Coupland & Bishop, 2007; Sharma et al., 2022).

On the other hand, while the Irish sea separates Northern Ireland from Scotland, the north of England and Wales, and much greater geographical distance separates it from the south of England, the ubiquitousness of SSBE as the language of politics, media and culture in the UK (e.g. Agha 2003) means that speakers of Belfast English are likely to be familiar with SSBE to a much greater extent⁶. Indeed, Northern Irish speakers rate SSBE as highly prestigious (Coupland & Bishop, 2007; Sharma et al., 2022). Northern Irish speakers also consider a standard Northern Irish English variety to be somewhat prestigious. However, in terms of social attractiveness, SSBE scores low for Northern Irish English speakers, while their judgment of the social attractiveness of Northern Irish English is somewhat mixed: Coupland & Bishop (2007) find strong in-group loyalty with Northern Irish raters, but these results are not replicated to the level of significance in Sharma et al. (2022)⁷.

Finally, it is worth noting that these judgments on prestige and attractiveness are highly mediated by age—or, attitudinal changes across the lifespan. Across the UK, speakers in middle age (i.e. the population that the participants in this study belong to) rate regional accents, including Northern Irish English, lower than younger speakers do, and place more value in the prestige associated with the SSBE variety (Sharma et al., 2022).

For our study, then, the social relationship between the L1a and the L1b is not straightforward. The SSBE speaker does not necessarily *require* accommodation or acquisition of new features to become established in the Belfast speech community—their L1a variety is familiar to their interlocutors, intelligible, and carries a level of prestige. However, to be considered a member of the in-group, or be seen as more socially attractive, it may be advantageous to acquire some features of Belfast English. This is quite different to the case of a Belfast English speaker who migrates to an

⁶ L1 Belfast English speakers may be bidialectal in a form of English which more closely fits with SSBE morphosyntactic patterns, and style shift according to their audience. To what extent this occurs may be mediated by a variety of social factors: see Milroy (1984).

⁷ Perceptions of accent in Northern Ireland may be further mediated by social, political and religious affiliations. See Corrigan (2010:25-27) for an overview of relevant factors.

SSBE environment, where their variety is likely to be unfamiliar, leading to potential issues around intelligibility, as well as a lack of prestige and social attractiveness driving a pull to accommodation and/or register shift into an existing, more ‘standard’ variety. We propose that this sociolinguistic dynamic makes the L1a SSBE population in Belfast a good candidate for investigating attrition as a result of changed input for adult speakers, given that any changes to the grammar exhibited are unlikely to be as a result of accommodation or dialect levelling towards the standard in a population that is already bidialectal.

3.4 Variants under investigation

3.4.1 Northern Subject Rule

The Northern Subject Rule (NSR) is one instantiation of the variation in availability of agreement with *-s* which is seen in many non-standard Englishes. Under the NSR, third person plural contexts exhibit variation, with both the standard *-ø* agreement (2a) and *-s* agreement (2b) available. However, *-s* is only permitted when the subject is a full DP, and not with the third person plural pronominal (2c). The agreement patterns can also be seen with auxiliaries, such as *be* or *have*, which do not exhibit standard agreement marking (2d-e).

- (2)(a) These cars go very fast.
- (b) These cars goes very fast.
- (c) *They goes very fast.
- (d) The eggs are/is cracked.
- (e) *They is cracked. (Belfast English, Henry 1995:16)

The NSR is reported as a feature of non-standard Englishes in Scotland (e.g. Adger & Smith 2010, Murray 1873, Pietsch 2005), Tyneside (e.g. Buchstaller et al. 2013), and Northern Irish Englishes, including Belfast (e.g. Corrigan 2010, Finlay 1994, Henry 1995, Hickey 2007, McCafferty 2003). Reports of usage rates in Belfast English and the surrounding varieties vary: Pietsch (2005) finds a reduction from 50% of non-standard *-s* agreement with third person plural DP subjects in speakers aged 65-75 to 30% in speakers aged 35-45, recorded in the 1970s. However, Henry (2002) finds 55% use of NSR by adult speakers in her caregiver and child corpus recorded in 1998, and Corrigan (2010:59) provides numerous examples of NSR agreement from speakers across age groups, recorded in 2009. In terms of its syntax, while some accounts (e.g. Henry 1995, Tortora & den Dikken 2010) analyse the variability using multiple subject positions, we follow Adger & Smith’s (2010) account of the NSR in Buckie Scots⁸. Different features and feature bundles on the functional head T (Tense)

⁸ Adger & Smith (2010) do not explicitly state that their account covers the data from Belfast, on the basis of claims from Henry (1995) about the lack of availability of NSR in interrogatives. However, NSR in interrogatives is widely recorded in Northern Irish varieties (Montgomery 2006) and is the most frequent type of interrogative agreement found in the rural dialects surrounding Belfast (Pietsch 2005). Henry (2005) finds some

map to different morphophonological spell outs, deriving a Pool of Variants for any given agreement context from which one will be selected for production based on a complex function including structural and phonological properties, recency, priming, and social factors such as age, class and interlocutor. For a variety with NSR, the required mappings are (see Adger & Smith 2010:1124 for the full derivation):

- (3) i. [singular:+, participant: -] $\rightarrow$ -s
- ii. [singular:-] $\rightarrow$ - $\emptyset$
- iii. [participant:+] $\rightarrow$ - $\emptyset$
- iv. [pronominal: -] $\rightarrow$ -s

These feature mappings account for the data in (4a-c) by deriving 100% - $\emptyset$ inflection for a case like (4c) with a pronominal subject, but a 50/50 split between - $\emptyset$ and -s for a full DP subject.

- (4)(a) T[participant: -, singular: -, pronominal:+] (e.g. *they*) – rule (ii) $\rightarrow$ - $\emptyset$.
- (b) T[participant: -, singular: -, pronominal: -] (e.g. *the cars*) – rule (ii) or (iv) $\rightarrow$ - $\emptyset$ or -s.

Under the predictions of the AvA model, then, this property should be amenable to attrition for our SSBE speakers: while their grammar does not have the [+/- pronominal] feature on T, the distinction between pronominals and full DPs is a feature of the variety, and thus reassembly of the feature mapping to allow this feature to participate in agreement should be possible. Combined with the closely related nature of the two varieties allowing for input to regularly become intake, we predict that attrition—in the sense of an acquired change to the L1 grammar giving rise to the agreement pattern of the NSR—is likely for the L1a SSBE speakers in Belfast.

3.4.2 Embedded inverted questions

In SSBE, while matrix interrogatives exhibit subject-verb inversion, embedded questions do not. Instead, in polar contexts, they add an overt complementiser (*if/whether*) (5a-b)⁹.

- (5)(a) Is she coming?
- (b) I asked **if/whether** she was coming.

Matrix interrogatives in Belfast English operate in the same way as (5a), and embedded interrogatives with an overt complementiser, like (5b) are also available. However, as well as this, subject-verb inversion is also possible in embedded questions in this variety, and in these cases there is no overt

evidence of NSR in polar questions in Belfast, and there is evidence of NSR in X (Twitter) data from Belfast (e.g. ‘Why is the girls not doing a meet and greet for Belfast?’ (@christopherpa20, 25 October 2012), ‘Does the boys need a security guard for Belfast?’ (@erinmccannn, 5 June 2014). We thus assume that the NSR *is* available in interrogatives in Belfast English, and that Adger & Smith’s analysis therefore extends to the variety.
⁹ We restrict discussion to embedded polar questions, but note that the relevant phenomena are also exhibited in *wh*-questions.

complementiser, such as in (6a-b). EIQs are restricted to non-factive, ‘rogative’ verbs, such as *ask* and *wonder*, although factive predicates (e.g. *know*, *remember*) can be licensed in special cases where the matrix clause is negative or interrogative (e.g. 6c)¹⁰.

(6)(a) I asked Jack was she in his class. (McCloskey 2006:88)

(b) this guy said he liked me and asked could he take me out on a date

(c) I don’t know is it damage. (Corrigan 2010:69)

Embedded inverted questions (EIQs) like (6a-c) are attested in Hiberno Englishes, including Belfast English (e.g. Henry 1995, Corrigan 1997, McCloskey 2006, Fippula 1999, Hickey 2007, Corrigan 2020). They are also found in some other Englishes, including in the north of England (e.g. Woods 2016), African American English (e.g. Green 2002) and Indian English (e.g. Stringer 2015), but crucially not in SSBE. Corrigan (2010) finds the structure is widely used in the Northern Irish and Belfast English corpora she investigates; it appears to be a robust feature of the variety.

In terms of its syntax, McCloskey (2006) proposes that all questions embedded under rogative verbs are recursive CP constructions with two C heads. C₁ is the higher of the two which is selected for by the verb. It has a [Q] feature, and spells out as the overt complementiser *if*. The lower C₂ has the feature bundle [Q, uT]. This cannot be selected by a higher lexical head, and so can only be in the root context or C₂ in varieties that permit EIQs, such as Belfast English. In SSBE, C₂ would be empty, but the relevant feature bundle would exist in matrix questions and so feature reassembly should be possible. However, Woods (2020) shows that EIQs have different syntactic and semantic behaviour, as well as pragmatic properties, that distinguish them from indirect questions with overt complementisers—such as the effects of negation and modality which make EIQs acceptable with non-rogative verbs (6c). Retaining the idea of the recursive CP analysis, she proposes that the EIQ is not the complement of the matrix predicate but rather an appositive relative clause in the style of Potts (2005). The lexical verb (e.g. *ask*) selects for a (silent) nominal complement which forms a constituent with the EIQ via a parenthetical head (Griffiths & de Vries 2013). Rather than a simple task of feature reassembly, then, an SSBE speaker in Belfast would need to acquire the silent nominal complement to allow the inversion to be embedded (see also Woods 2016:211 on the distinction between EIQ and non-EIQ dialects). We thus predict that, although attrition—here, the acquisition of EIQs—is possible in this context due to the likelihood of input becoming intake in a situation of contact between two closely related varieties, it is less likely than in the context of the NSR as it is not a straightforward case of feature reassembly.

In the next section, we present the methodology and results of our data collection process.

¹⁰ The embedded interrogative presupposes an unresolved question in the discourse, which is incompatible with a resolute matrix predicate like *remember* or *know*. By negating or questioning the matrix predicate, the presumption of resolution is removed, and an open question can be embedded (McCloskey 2006).

4 The data

4.1 Methodology

4.1.1 Participants

The experimental group consisted of 30 L1a SSBE speakers (18 women, 12 men) who had moved to Belfast 15+ years ago. The average age of the experimental group was 51.3 (range: 35-71); the mean age of arrival in Belfast was 26.5 years (range: 19-39 years), and the mean length of residence in Belfast was 24.7 years (range: 15-51 years). We also collected data from a control group of 30 SSBE speakers who had lived their whole lives in an SSBE-speaking area (14 women, 16 men). To age match with the experimental group, the controls were aged between 30-69, with an average age of 48.

Experimental participants had moved to Belfast for a range of reasons: most commonly, due to meeting a partner who was from Northern Ireland, or for work or study. Participants completed a background questionnaire based on the sociolinguistic questionnaire for language attrition (Keijzer 2007, Schmid 2011) and the network density questions of Milroy (1980:141). Principal Components Analysis conducted on the questionnaire data indicates that three main components arise from the data: a participant's feelings of affiliation towards England; the strength of their Northern Irish network, and the frequency/length of their visits to England. We will draw on these factors where relevant in our discussion of the data below. For more information, please see Appendix A.

4.1.2 Instruments

The data we present in this paper comes primarily from an acceptability judgment task (AJT) which was administered to the participants online via Qualtrics, and facilitated over Microsoft Teams by an SSBE fieldworker also living in Belfast. Participants first read a short context which established relevant information. They then pressed a button to hear a speaker of SSBE produce a sentence which followed the context, and they read the sentence on screen. Participants were then asked to rate the acceptability of the sentence on a Likert scale from 1-5, where 1 indicated 'I would never say that, it's very unnatural, it's bad' and 5 indicated 'I would definitely say that, it's very natural, it's fine'. For the NSR, participants saw 24 contexts twice, once with the standard agreement and once with -s agreement, totalling 48 examples (7a-b) (see Table 1 for the experimental design). For EIQs, participants also saw 24 contexts twice, once with a standard *if* complementiser¹¹, and once with embedded inversion, totalling 48 examples (8a-b) (see Table 2 for the experimental design).

¹¹ *If* is more frequently used than *whether* across English varieties (e.g. Biber et al. 1999, Kolbe-Hanna 2021) and so we chose *if* for our materials.

- (7) I ask why you couldn't make it to the cinema at the weekend. You say:
- (a) My friends was having a big party. [NSR agreement]
- (b) My friends were having a big party. [standard agreement]
- (8) You tell me about a conversation you had with some colleagues about France. You say:
- (a) I asked had they been to Paris. [embedded question with inversion order]
- (b) I asked if they had been to Paris. [embedded question with standard order]

Test items were pseudorandomised alongside 48 filler items, which exemplified other non-standard features such as negative concord and *have*-raising. In total, then, participants each judged 144 items and the task took around 30 minutes.

	- $\emptyset$ agreement	-s agreement ¹²
3 rd person plural DP	e.g. My friends were having a big party.	e.g. My friends was having a big party.
3 rd person plural pronominal	e.g. They were having a big party	e.g. *They was having a big party.

Table 1: Experimental design for the NSR condition, indicating the expected judgments if the participants have undergone grammatical attrition of this property

		<i>if</i>	EIQ
Regular declarative matrix	Rogative verb	e.g. I asked if they had been to Paris.	e.g. I asked had they been to Paris.
	Non-rogative verb	e.g. I knew if they had been to Paris.	e.g. *I knew had they been to Paris.
Negative declarative or interrogative matrix	Rogative verb	e.g. I didn't ask if they had been to Paris.	e.g. *I didn't ask had they been to Paris.
	Non-rogative verb	e.g. I didn't know if they had been to Paris.	e.g. ?I didn't know had they been to Paris.

Table 2: Experimental design for the EIQ condition, indicating the expected judgments if the participants have undergone grammatical attrition of this property

Participants also took part in a 30-60 minute sociolinguistic interview (e.g. Labov 1972a) over Microsoft Teams with the SSBE fieldworker, on topics such as their life history, their work, and their move to Belfast. Having interviewers who were in the same linguistic position as the experimental

¹² In the initial design, verbs used were subset into *was/were* and 'other verbs', which included *come*, *go*, *have* and *do*, as *was/were* variation and particularly *was*-levelling is a broader phenomenon across subject forms in non-standard varieties of English (e.g. Britain 2002, Cheshire & Fox 2009, Smith & Tagliamonte 1998, Tagliamonte 1998). However, there was no effect of verb type on any of the group results, and so we combine them for the purposes of analysis.

participant group, e.g. L1a SSBE speakers who received regular and extensive input in Belfast English and shared relevant community insight with the participants, ensures as far as possible that the context is set for evidence of *long-term accommodation* (or, grammatical attrition) rather than any effects of the Observer’s Paradox leading to accommodation towards the fieldworker. The data from the interviews, totalling approximately 20 hours and 208,000 words of potentially attrited speech, was fully transcribed and anonymised, and is freely available as part of the Corpus of Native Grammar Attrition (CoNGA) (conga.soton.ac.uk). Where relevant, the judgment data will be compared with examples from the corpus.

4.2 Northern Subject Rule – results

Attrition of the NSR for our L1a SSBE experimental group would result in higher acceptability of non-standard *-s* agreement with third person plural DPs, but rejection of *-s* agreement when the third person plural subject was pronominal. We treat the data as non-linear and thus report medians and modes rather than means throughout.

Looking at Figure 2, it is clear there is no evidence of attrition for the experimental participants, with each point representing the median for an individual¹³.

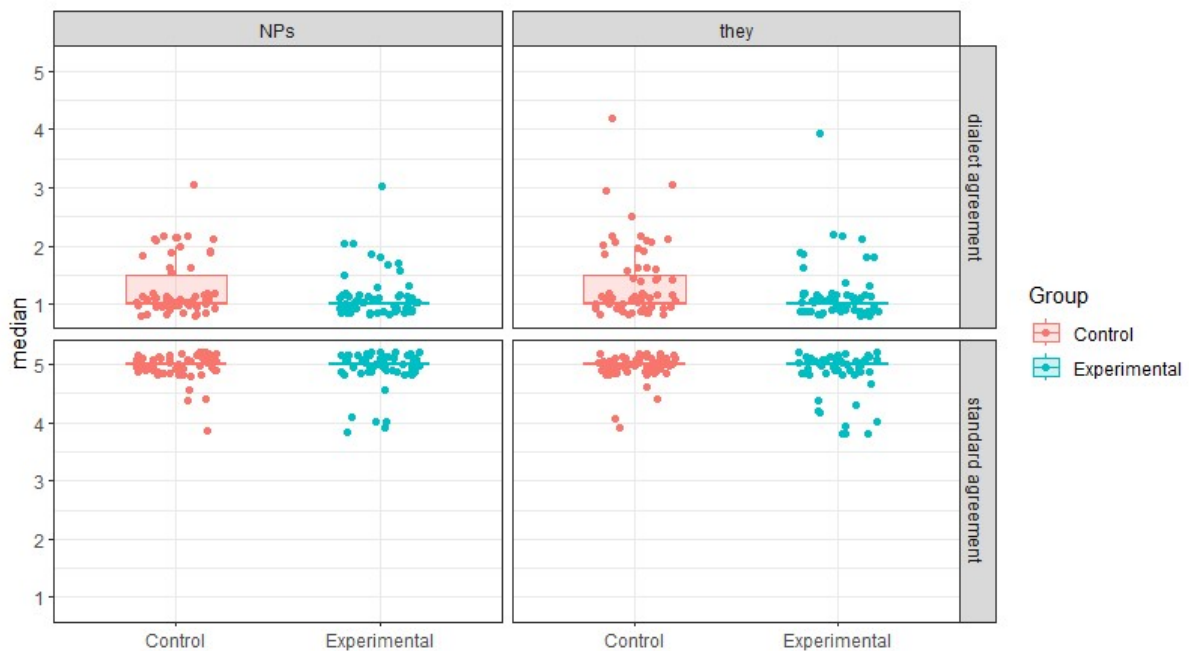


Figure 2: Median ratings of acceptability for standard and non-standard agreement

¹³ We note that there are two outliers, one in each group, who have a particularly high median rating of 4 for the non-standard *-s* agreement in the pronominal subject context. These ratings arise from examples with *was*. *Was*-levelling is found in areas where SSBE is spoken; for example, Cheshire & Fox (2009:11) report over 40% use of *was* with *they* in some of their London speaker groups. Acceptance of this form may thus derive from *was*-levelling in the L1a.

Standard - $\emptyset$ agreement received a median and mode rating of 5 from participants in both the experimental and control groups, with both pronominal and full DP subjects. When considering -s agreement, the NSR was not apparent in participants' judgments, with both pronominal and full DP subjects receiving median and mode ratings of 1. We ran a cumulative link mixed model (CLMM) in R (Christensen 2022) to analyse the group level results. CLMMs are forms of ordinal logistic regression designed specifically for dealing with multiple, discrete response categories, such as the judgments given in this study. We find a main effect of the type of agreement (whether it is -s agreement or the standard - $\emptyset$ agreement), but no effect of or interaction with the type of subject (pronominal or DP). Furthermore, there is no effect of group (experiment or control), and within the experimental group there are no effects of the factors identified in the Principal Components Analysis.

Coefficients	Estimate	SE	z-value	P value	
Agreement -s	-8.87	0.33	-26.62	< .00001	***
Observations: 2829. Random effects: EXAMPLE (48, s.d. 0.75), PARTICIPANT (59, s.d. 1.23)					

Table 3: Results of CLMM for NSR AJT

We extracted all possible contexts for use of the NSR by participants in their sociolinguistic interviews. However, no use of the NSR by participants was found.

4.3 Embedded inverted questions – results

Attrition in the EIQ context would result in greater acceptance of EIQs under a rogative predicate (e.g. *ask*, *wonder*) in a matrix declarative construction (9a) and under a non-rogative predicate (e.g. *know*, *find out*) in either a question or negative declarative structure (9b)¹⁴¹⁵.

(9)(a) I asked had they been to Paris / I asked if they had been to Paris.

(b) Did she know was it going to close? / Did she know if it was going to close?

The results can be seen in Figure 3, where each point represents the median for an individual¹⁶.

¹⁴ Both questions and negative declarative structures were included as the mechanism which permits embedded inversion in these constructions is the same—the cancellation of the ‘resolution’ presupposed by the matrix predicate.

¹⁵ EIQs under non-rogative predicates in matrix declarative constructions are generally rejected by EIQ users, though they are reported in limited circumstances in Belfast English by Henry (1995) and Corrigan (1997).

¹⁶ We note the large variation in responses to *if* under non-rogative verbs (e.g., *know*). *Know* is a responsive verb which can, theoretically, embed either declarative or interrogative complements. Many semantic theories of responsive verbs propose that regardless of the interpretation of the complement, the predicate is a single lexical item which selects for propositions, not questions (e.g., Heim 1994, Dayal 1996, Lahiri 2002, Spector & Egré 2015). It may be that the semantics of the particular contexts we tested led to preferences for declarative interpretations to be upheld and thus slight dispreference for the interrogative complementiser *if*, where *that* would also be possible. We set this to the side as it is not relevant for our discussions of attrition.

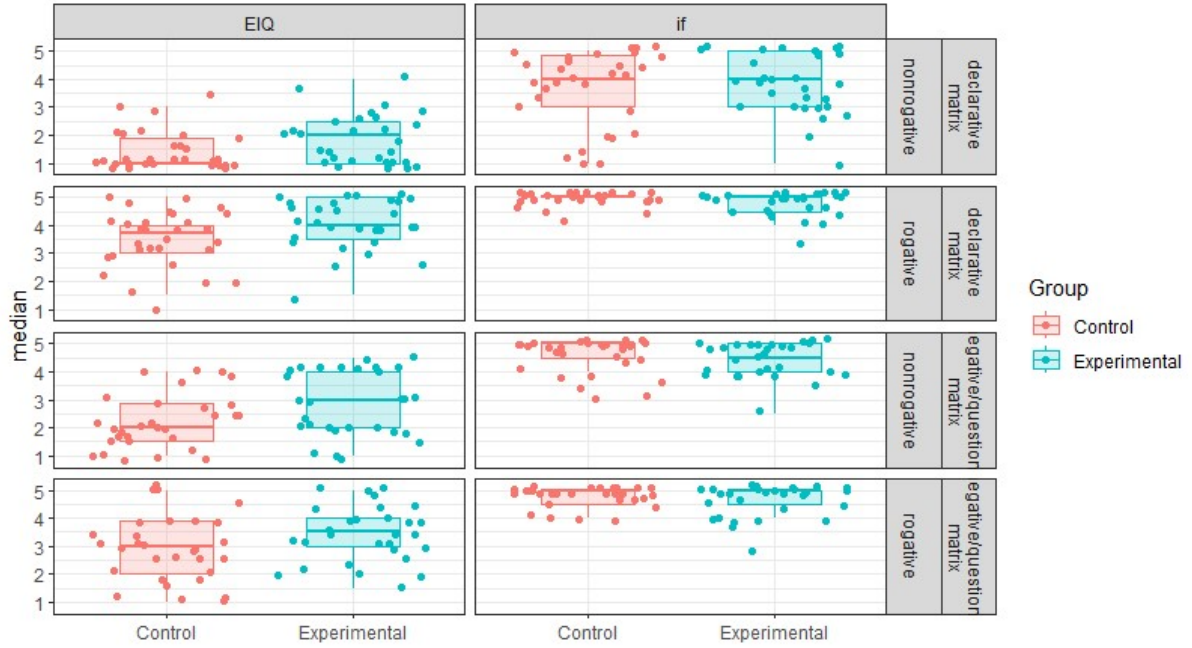


Figure 3: Median ratings of acceptability for EIQs and overt complementiser *if*

As the predictions were different for the two matrix clause contexts, we subset the data for analysis. In the descriptive statistics, there are some differences between the groups in the acceptability of EIQs. For EIQs under rogative verbs in declarative matrix clauses (the second row of Figure 3), the experimental group has a median of 4 and mode of 5, while the control group has a median and mode of 4. There is a somewhat higher acceptability of the construction for our participants in Belfast, although the control participants do not strongly reject it. We do also see higher acceptability of EIQs in general under negative/interrogative matrix clauses (the bottom two rows of Figure 3) for the experimental participants (median 3 under both rogative and non-rogative verbs, mode 5 under rogative verbs and 2 under non-rogative verbs) than for the controls (median and mode 3 for rogative verbs and median and mode 2 for non-rogative verbs). In term of the predictions, then, it seems as though there may be greater acceptability of the EIQ under rogative verbs with a declarative matrix, but the prediction of greater acceptability of EIQs with non-rogative verbs under negative/interrogative predicates is not borne out.

We ran CLMMs in R (Christensen 2022) to further analyse the results.

Coefficients	Estimate	SE	z-value	P value	
Q-type	2.93	0.29	9.98	< .00001	***
Matrix verb type	3.12	0.30	10.52	< .00001	***
Observations: 1416. Random effects: EXAMPLE (24, s.d. 0.64), PARTICIPANT (59, s.d. 1.18)					

Table 4: Results of CLMM for embedded questions with declarative matrix clauses

For examples with a declarative matrix clause (9a), we find that there is an effect of construction type (questions with *if* are more acceptable than EIQs) and an effect of main verb type (rogative verbs are more acceptable than non-rogative verbs). However, there is no interaction between the two effects, and there is no effect of group—i.e. despite differences in the descriptive statistics, there is no difference between the experimental and the control groups in how they judge EIQs.

Coefficients	Estimate	SE	z-value	P value	
Q-type	3.55	0.36	9.85	< .00001	***
Matrix verb type	0.85	0.35	2.43	< .015	*
Observations: 1413. Random effects: EXAMPLE (24, s.d. 0.81), PARTICIPANT (59, s.d. 1.22)					

Table 5: Results of CLMM for embedded questions with negative/interrogative matrix clauses

For examples with a negative or interrogative matrix clause (9b), we find that there is an effect of construction type (questions with *if* are more acceptable than EIQs) and an effect of main verb type (rogative verbs are more acceptable than non-rogative verbs). However, there is no interaction between the two effects, and there is no effect of group—i.e., there is no difference between the experimental participants and the control group in how they judge EIQs.

At group level, therefore, we do not see any evidence of attrition, in terms of acquisition of EIQs by the L1a SSBE speakers in Belfast. However, as we saw in the descriptive statistics, there is variation in the behaviour of individuals in their acceptability, particularly with the rogative verbs in declarative matrix clauses. Attrition is notoriously subject to individual variation (e.g. Köpke 2007); it is worth checking if there are particular speakers who have more categorical judgments.

For declarative matrix clauses with rogative verbs, there are 9 experimental participants who have a median and mode of 5, and 4 of those participants gave a rating of 5 to all the relevant examples. There are only 3 members of the control group who also have a median and mode of 5 for this construction, and none of them rated all relevant examples as 5. The fact that there are such a number of experimental participants who rate these constructions highly is particularly notable, as in general, participants who are in a situation of linguistic change or instability usually reject more strongly than control groups who are not faced with linguistic uncertainty (Jamieson 2020, Orfitelli & Polinsky 2017, Hicks et al. 2024). We note that there are no apparent sociolinguistic factors linking this group of participants: their length of residence range from 15-25 years, and while the majority have partners from Northern Ireland, not all do. They do not group together in their scores for any of the factors in the Principal Components Analysis—i.e., they do not report especially tight-knit Northern Irish networks compared to the rest of the participants, nor do they report lesser affiliation to England than the participants overall. We therefore simply note that there does appear to be greater

acceptance of EIQs in their most canonical context for some participants, even though we do not see attrition at group level confirmed by the quantitative results.

Finally, to look at the corpus data, we find two examples of EIQs produced by L1a SSBE speakers now living in Belfast, both of which are *wh*-questions embedded under rogative predicates in declarative constructions (10a-b).

(10)(a) People here ask where are you from. [BP13]

(b) I wonder what will ha- what will I do when he's at school. [BP14]

The first example was produced by BP13, a 46-year-old woman who is originally from Southampton, and had been living in Belfast for 19 years. The second example was produced by BP14, a 53-year-old man originally from Nottingham, living in Belfast for 30 years. Neither of these L1a regions are reported as having EIQs, and so we assume that the production is due to some influence from Belfast English. Both have Northern Irish partners, and both have primarily Northern Irish self-reported networks, though BP14's is notably more tight-knit. Comparing the corpus data with the judgment data, neither are in the group discussed above who categorically or especially strongly accept the EIQ construction, though they do both rate the examples highly (BP13: median 4, mode 4; BP14: median 4.5, mode 5). Again, we note that though there was no group-level quantitative attrition, individual participants do appear to have undergone some attrition (i.e. acquired EIQs).

4.4 Summary

We predicted, based on the AvA model discussed in section 2, that grammatical attrition would be likely for L1a SSBE speakers exposed to L1b Belfast English due to the closely related nature of the two varieties. Furthermore, we predicted that attrition would be more likely with respect to the NSR, as to acquire the NSR involves a more straightforward feature reassembly task than acquiring EIQs. However, our results do not bear out these hypotheses. We found no evidence of attrition for the NSR in the data, with participants rejecting it completely; little individual variation, and no attestations in the corpus. On the other hand, while we found no evidence of group-level attrition for the EIQs, there is considerable individual variation, including ceiling-level acceptance for a number of experimental participants and production in the corpus. This variation does not clearly map to any expected sociolinguistic predictors, such as strength of Northern Irish network or affiliation towards England.

5 Discussion

5.1 Assessing the AvA model

In section 2, we outlined where there was convergence between models of morphosyntactic change in both language attrition and variationist sociolinguistic conceptions of long-term accommodation. We argued that a formal model of grammatical attrition like Hicks & Domínguez's AvA model allowed us

to make clearer predictions about for which linguistic pairings and for which morphosyntactic features change was likely to occur. How well did the predictions made by the model account for the results we saw in section 4?

Firstly, in terms of the *perceptual encoding* module of the model, it was predicted that attrition would be likely. Due to the overall similarity of the two varieties as relatively close dialects of the same L1, converting input to intake should be a comparatively easy feat; processing of the L1b input should be sufficiently successful that specific grammatical differences in the input are detected. There may be some early challenges with features of the accent, lexical items, or discourse-pragmatic phenomena, but for the most part, the SSBE speakers would not need to actively acquire a new variety in order to understand speakers in the community, and given the widespread availability and prestige of SSBE in Belfast, even less need to do so to be understood. Furthermore, the length of residence of the experimental participants in this study—on average, 24.7 years—means their exposure to the L1b has been extensive. They have had plenty of opportunity to hear the relevant features, and thus convert the input to intake. Both of these facets of the migration experience have been claimed to be important for potential change in both the attrition and the variationist literature. However, despite these general features of the perceptual encoding model holding, we did not see an increased acceptance of the L1b structures at group level.

We therefore turn to the second module in the AvA model—the *inference engine*, and the possibility of converting the perceptual intake to acquisitional intake. Again, here, the predictions of the model were not exactly borne out. While attrition was predicted to be possible for both properties at the perceptual encoding stage, in the inference engine they diverged given the implementation of feature reassembly proposed. The hypothesis was that attrition would be more likely in the case of the NSR, for which feature reassembly is more straightforward with the utilisation of a [+/- pronominal] feature within the agreement paradigm, following Adger & Smith (2010). Attrition was proposed to be less likely for EIQs, where, following Woods (2020), SSBE speakers in Belfast needed to acquire a silent nominal complement, and the ability to build the appropriate parenthetical structure to invert the question. However, what we see is evidence of individual-level attrition in the form of some acceptance and acquisition of EIQs, and no evidence of attrition for the NSR.

One possibility is that the analyses of the constructions we followed in building the hypotheses were not the correct ones. Following the analysis of EIQs presented in McCloskey (2006) would make the proposed feature reassembly task a more straightforward one for the SSBE speakers; speakers would have to acquire the possibility of rogative verbs selecting for a C₂ which was specified with [Q, uT]—a feature bundle which already exists on their root C. The feature reassembly task would thus be a straightforward move of the existing bundled C head into the embedded C₂. If we were to follow this analysis of EIQs, it is hard to say which of the properties would be more amenable

to attrition, as both would involve a straightforward feature reassembly task. A change to the underlying analysis of the properties in question in itself thus does not account for the result of some attrition in the EIQs and none in the NSR, but it indicates that a clear understanding of the feature reassembly task at hand is essential for making predictions within formal models of attrition in order to better understand where change may occur.

In summary, while the AvA model allows us to make clear predictions about where morphosyntactic attrition may occur for an adult speaker of one dialect exposed to extensive input from a second dialect, as it stands, the model alone cannot account for the results of the study we see above. In the next section, we discuss how insights from variationist theory can be used to further specify the AvA model and help us develop a clearer picture moving forward of how to understand grammatical change as a result of long-term exposure to a second dialect.

5.2 Further specifying the AvA model with variationist theory

The AvA model, and language attrition models in general, take it as given that once a speaker can perceptually encode the input that they are receiving in their second language or dialect, it can be converted to intake, and thus grammatical attrition can emerge. However, as we saw in this study, even when input can straightforwardly become intake, with no issues of mutual intelligibility, attrition may not occur at group level, or at all. Further specification of the model is required. We believe this specification can readily come from conclusions reached about dialect contact in the variationist sociolinguistic framework. Specifically, we consider how *salience* may affect possible perceptual encoding of the relevant properties. We then consider how the model could be updated to consider individual speakers' *volition* to change as accessible perceptual encoding but suppression of the inference engine. Finally, we discuss the possible role of variation/*optionality* in the original dialect grammar as a context that may favour change and how this may help further specify the inference engine module.

5.2.1 Salience

Variationist dialect contact literature situates salience centrally in its understanding of morphosyntactic change, with salience defined as some property of the linguistic feature that makes it prominent in perception for speakers. Pinpointing exactly what salience is can be difficult, and varies from feature to feature on both language internal and external dimensions. Language-internal factors include phonological prominence, the reliability of a form-meaning mapping, and frequency, while language-external factors cover things like social stigma. Whatever the reason for the salience of a particular linguistic feature, it appears salient features are more vulnerable to potential changes (Auer et al. 1998, Kerswill & Williams 2002a, Trudgill 1986), with the caveat that this is particularly strong in the case where change amounts to adding a new property, while in the case of loss of a property, the

effects of objective language-internal salience (e.g. phonological factors and lexicalisation) are mediated by attitudinal factors¹⁷.

Salience can help account for the results seen in section 4, considering both perceptual (phonological) salience and semantic reliability. Turning firstly to the NSR, it is well-documented in both child language acquisition (since Brown 1973) and adult second language acquisition (see Goldschneider & DeKeyser 2001 for an overview) that verbal agreement with *-s* in English is acquired later than other verbal morphemes (such as progressive *-ing*), particularly as a cue for comprehension. Lack of perceptual (phonological) salience is proposed to be a factor in the challenge to acquire the property: for example, Legendre et al. (2014:45) argue that the C_# environment which generally embeds third person singular *-s* in English is not conducive to phonological salience, given that consonants are perceived more clearly in pre-/inter-vocalic environments.

This lack of perceptual salience is further enhanced by evidence from processing studies with native (standard) English speakers. When presented with written instances of ‘violations’ of subject-verb agreement in English (i.e. examples of NSR), P600 effects are found (Coulson et al. 1998, Kaan et al. 2000, Osterhout & Mobley 1995). However, when using auditory stimuli, Dube et al. (2016) find that for the Australian English speakers in their sample, there is a P600 effect when the incorrectly marked verb is in utterance-final position, but effects are ameliorated when the verb is utterance-medial. Dube et al. propose that when in utterance-final position, there is increased perceptual salience of the agreement violation due to phrase-final lengthening, which allows for listeners to notice the mismatched agreement form. Extending this to our SSBE participants in Belfast, there is no standardised or common orthography of Belfast English, and the absolute majority of input participants will receive will come from *hearing* the dialect. In non-experimental speech contexts, instances of NSR will be far more frequent in utterance-medial contexts—for example, Hsieh et al. (1999) find that only 16% of English third person singular *-s* was produced in sentence-final contexts. We thus propose that the NSR may not be phonologically salient in the input for our SSBE speakers in Belfast.

Turning to semantic regularity, it is expected that if there is a clear form-meaning mapping between a morpheme and its semantic output, it will be salient (Brown 1973, Mufwene 1991). English third person verbal *-s* does not fit this bill, even in the standard variety. Firstly, it is homophonous with both plural and possessive *-s*, and secondly it marks number, person *and* tense. This level of complexity has been proposed as another factor in the difficulty of acquiring the marker (Brown 1973, Goldschneider & DeKeyser 2001). In the case of our SSBE speakers in Belfast, the NSR presents an

¹⁷ It is possible for features to be ‘too’ salient; features which serve as *stereotypes* (Labov 1972b) of a particular dialect may be avoided completely in accommodation (Trudgill 1986). Neither of the properties investigated here falls into this category.

even more complex system of form-meaning mappings than their existing verbal agreement system. While their *-s* agreement indicates a present tense verb in the third person singular, the NSR adds an additional form-meaning mapping of number to *-s* with the third person plural—but only if the subject is non-pronominal. The lack of clear semantic function for the *-s* morpheme in the input, combined with its low phonological prominence, means that it likely flies under the radar for SSBE speakers in Belfast—even after an extensive period of residence and high levels of input. The NSR is thus unlikely to be perceptually encoded for adult SSBE speakers receiving Belfast English input, and is thus unable to become intake and feed through to the inference engine. The ease of the feature reassembly task is thus irrelevant for the NSR case.

On the other hand, there is reason to believe that EIQs may be more salient for SSBE speakers¹⁸. In terms of phonological salience, there are major differences between an embedded question with *if* and an EIQ. Firstly, the loss of a non-prominent, single syllable lexical item *if*, likely to be phonologically merged with the preceding verb (i.e. *asked if* /ask tɪf/). This is replaced in the sequence by a full verb e.g. *did*, *was* which will not phonologically merge with the matrix verb, and is likely to have greater prosodic prominence within the structure (see Gleitman et al. 1984 on the salience of auxiliaries in first position in child language acquisition). There is then also a gap following the subject where the auxiliary would be expected to be. Phonology alone thus makes the EIQ more likely to be salient in the input for our SSBE speakers.

In terms of semantic complexity, despite their similarities EIQs are understood to have a different semantics to non-inverted embedded questions. EIQs have an intuitively closer link to direct speech, and analyses (McCloskey 2006, Woods 2016 - see section 3.4.2) have proposed that they are a form of interrogative speech act, previously considered to be exclusively a matrix phenomenon. This type of construction is not available in dialects of English which do not have EIQs, like SSBE. Therefore, rather than adding a new mapping to an existing form (as we saw in the case of the NSR) or incorporating a new form that maps to an existing meaning (as would be the case if EIQs were truly indirect questions), for SSBE speakers EIQs are a new form with a distinct meaning. Furthermore, Cheshire (1996) proposes that contexts with greater interactional import (such as negation and interrogatives) are more likely to be salient for listeners, as ‘syntactic environments that require speaker-addressee collaboration in the production of discourse are presumably important to speakers’ (1996:8). EIQs’ links to direct speech make them inherently more interactional than the purely indirect speech report of the non-inverted embedded question. These factors taken together suggest

¹⁸ The discussion we present here is relevant for polar EIQs, as these are constructions we tested in the AJT. The salience facts may differ for *wh*-EIQs. However, from the acquisition literature we note the general tendency to accurately acquire polar interrogatives (both matrix and embedded) before acquiring their counterpart *wh*-questions (e.g. Klima & Bellugi 1966, Pozzan & Quirk 2014). The two examples of production of EIQs in the relevant data were *wh*-EIQs, suggesting that for at least some SSBE speakers in Belfast, they have moved to what we might assume is the second stage of the EIQ acquisition process.

that EIQs are likely to be more salient in the input for SSBE speakers than the additional complexities being added to a single phonologically non-salient morpheme by the NSR, with a detectable difference in perceptual input supported by an interpretative distinction. Together, this allows for the EIQ to become perceptual intake, and thus the inference engine can be fired up for the feature reassembly task.

5.2.2 Volition

Salience further specifies the perceptual encoding module of the AvA model, and can help us account for the fact that there was no evidence of attrition for the NSR, but some evidence of attrition in the EIQs. However, we did not find group level change; instead, it was only in certain individuals. We know that EIQs *can* become intake for speakers, but it seems that for some speakers, they either a) fail to convert the EIQ to intake, or b) somehow suppress the inference engine when it comes to updating their grammar. In the introduction to their 2005 book, Hinskens, Auer & Kerswill write that ‘in the case of the convergence and divergence between dialects in contact (where mutual comprehensibility is not at stake), volition and more generally sociopsychological factors play the main role’ (Hinskens et al., 2005:40). Volition in this sense would lead to active suppression of the inference engine despite perceptual intake. The role of volition and identity in the outcomes of dialect contact is not uniformly accepted (e.g. Mufwene 2001, Trudgill 2008 and the associated responses), and even Labov, whose Martha’s Vineyard study has influenced much work on the role of social identity in linguistic change, states ‘we do not often find correlations between degrees of local identification and the progress of sound change’ (2001:191). Nevertheless, we consider whether volition may help us further understand the evidence of attrition for certain individuals in the SSBE data. We use the results of our Principal Components Analysis to approximate volition to change.

In Figure 4, the two strongest components (Affiliation to England and Northern Irish network) from the analysis are plotted, with each experimental participant plotted as an individual point. From this, if participant’s volition to change was facilitating any potential attrition (i.e., acceptability or production of EIQs), we might expect that participants in the top left quadrant would be most like to demonstrate attrition. These participants reported dense social networks, consisting primarily of people from Northern Ireland. They also reported feeling less affiliated to England (e.g., they report feeling more at home in Northern Irish culture, a lack of homesickness and no desire to move back to England, and not feeling a strong need to maintain their ‘English’ dialect). However, looking at the participants for whom we see production or categorical acceptance of the EIQs, no clear pattern emerges. The circled participants are the two who produced embedded inversion; the rectangles indicate participants who accepted every canonical EIQ example. The participants are spread across the graph, rather than clustered in the top left corner as we might expect.

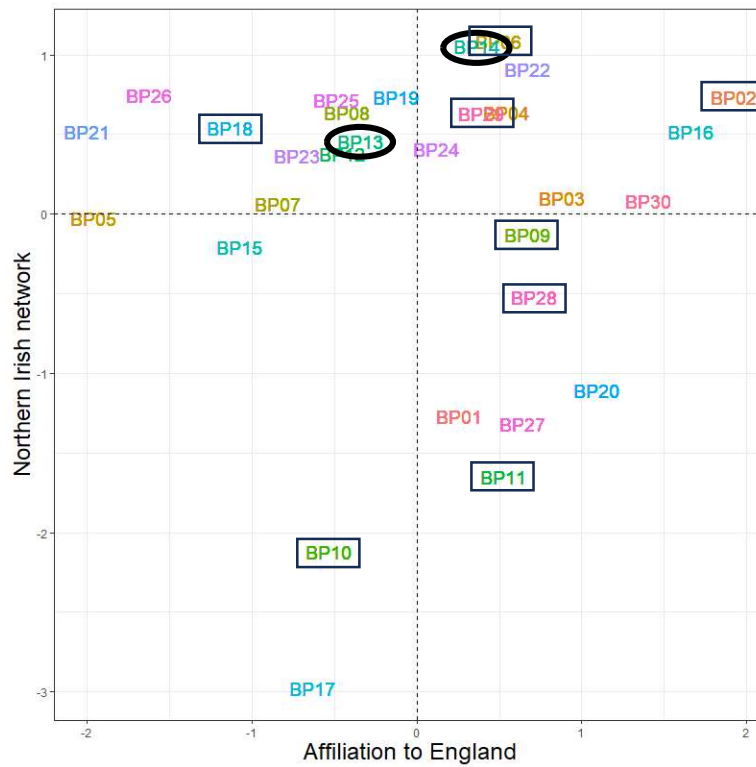


Figure 4: SSBE speakers in Belfast plotted according to strength/density of their Northern Irish network and the level of their continued affiliation to England

Although this result is, of course, not enough to rule out the role of volition in predicting changes to individual speakers' adult grammars as a result of extensive input from a new dialect, it seems as though we can set it aside in accounting for the results we see in this particular study. Further study could investigate whether volition and the ability to suppress change at the level of the inference engine is truly a driving force in individual-level change, or whether it is simply that some speakers are more sensitive to salience. If volition does turn out to be a relevant factor, a mechanism for suppressing the inference engine would be a require adaptation to the AvA model.

5.2.3 Existing optionality

While salience and volition help further hone the perceptual encoding and intake facets of the AvA model, we also wish to consider further specification to the inference engine. To this end, one final point where we believe sociolinguistic considerations may have an influence on grammatical attrition is the role that existing variation or optionality in the grammar of a first dialect may play. Furthermore, this ties to how we define attrition as opposed to accommodation in this paper. Recall that we focused on a particular case where any change to the grammar of the SSBE speakers would be *adding* a new option to their grammar, rather than increasing or decreasing their use of an existing feature. Research on accommodation more often focuses on speakers who are already bidialectal (e.g., in a local dialect and a standard variety), and how these speakers move towards the standard to be

understood. We leave a specific implementation of this more traditional form of accommodation within the AvA model for further research, though we note that the mechanisms should be similar.

Returning to the figures presented in section 4, for the NSR, there is very little variation in terms of how the feature is judged. Participants in both the control and experimental groups are fairly categorical in rejecting instances of non-standard agreement. While there is some agreement variation in southern varieties of English (namely, *was/were* levelling e.g. Britain 2002, Cheshire & Fox 2009), the constraints on the variation are very different in northern (and Northern Irish) varieties, with NP subjects dispreferred for *was* (i.e. the opposite of the NSR). Furthermore, the NSR applies to all verbs rather than a reflex of the irregularities of the *be* paradigm, which has fluctuated over centuries (Nevalainen 2006). There is, therefore, likely a very stable (and categorical) system of agreement in the SSBE speakers. On the other hand, in Figure 3 which shows the distribution of judgments for EIQs, there is considerable variation in the individual responses both for the control group and the experimental group, rather than any clear rejection. Woods (2016:63) notes that speakers of varieties of English which are not commonly understood as varieties with EIQs may, rarely, produce EIQs under *say* (11), despite rejecting them in judgment tasks.

(11) They said what did we want to be [. . .] I said a library lady.

(south west England, BNC cf. Woods 2016:63)

Clearly, this is not what we found in this study, as the only produced instances of EIQs were from SSBE speakers who had spent considerable time in Belfast. However, it may be the case that there is an underlying tendency towards the possibility of EIQs in the grammar of SSBE speakers which makes it more susceptible to change given extensive input from a variety which *does* robustly attest EIQs. While EIQs are not held to be ‘vernacular universals’ of English, we can certainly ask—are speakers of standard varieties who do not regularly attest features like negative concord (e.g., Blanchette & Lukyanenko 2019) or participle-preterite variation (e.g., Chatten et al. 2022) more likely to acquire these features than a purely dialectal feature? We set this aside as a question for future research.

5.2.4 Summary

We have shown that incorporating aspects of variationist theory into formal models of language attrition, specifically the AvA model, allows us to further specify how the various components of the model work, and better accounts for the data from SSBE speakers in Belfast presented in section 4. Salience, which has long been held as an important factor in the possibility of (long-term) accommodation in the variationist literature, provides greater specification of the perceptual encoding module of the AvA model. Factors affecting salience, such as semantic complexity, prosodic or phonological prominence, and issues around stigma and perceived identity associations with

particular linguistic features are all likely to play a role in determining when input becomes intake, and thus when attrition is possible. Incorporating this insight from the variationist framework also brings the notion of attrition as a form of acquisition closer to other types of acquisition, where salience has been noted to be an important factor. Volition to change, while not relevant to the results here, suggests that individual change could be due to a suppression of the inference engine *despite* the ability to perceptually encode input into intake for a particular feature of the grammar; a clear further direction for study lies in whether individuals who do not attrite for a particular feature suppress the processing of the intake, or whether they do not perceive the difference in the first place. Finally, the structure of the existing first dialect grammar and whether there is underlying optionality or variation within the grammar may impact the task of the inference engine, raising questions around the directionality of the change (i.e., accommodation of an existing feature vs. acquisition of a new feature) and the role played by phenomena like vernacular universals in enabling attrition. While we have not answered this question here, again we set out clear direction for future study, bringing together key factors from variationist models of grammatical change and models of grammatical change in attrition/acquisition frameworks to better understand the variable/individual nature of grammatical change and where it arises.

6 Conclusions

As cited in the introduction, Chambers (1992:703) states ‘if the principles of dialect acquisition gain empirical depth, their relationship to principles of first-language acquisition and second-language acquisition could open up a broad area of interaction among language researchers who have hitherto seldom recognized their common ground’. In this paper, we have taken up Chambers’ suggestion, and shown where and how language attrition and variationist sociolinguistics align in their assumptions about the outcomes of dialect contact in the first generation. Focusing on a case study of SSBE speakers who have been living in Belfast for decades, we showed how models from language attrition (specifically, the AvA model) can formalise assumptions in sociolinguistics about the role of input in accommodation as well as making predictions about the featural configurations of morphosyntactic properties likely to be acquired. We then explored how theories from sociolinguistics—salience, volition and underlying optionality—can be used to further specify the models and enrich discussion in language attrition about how input can become intake. We hope that we have shown there is much to be gained from bringing together perspectives from both fields, and this paper will thus act as a starting point for further discussion and incorporation.

References

- Adger, D. (2006). Combinatorial variability. *Journal of Linguistics*, 42(3), 503–530.
- Adger, D., Jamieson, E., & Smith, J. (in press). Sociolinguistics and minimalist syntax. In K. Grohmann & E. Leivada (Eds.), *Cambridge Handbook of Minimalism and its Applications*. Cambridge University Press.
- Adger, D., & Smith, J. (2010). Variation in agreement: A lexical feature-based approach. *Lingua*, 120(5), 1109–1134.
- Anderwald, L. (2005). Negative Concord in British Dialects. In Y. Iyeiri (Ed.), *Aspects of English Negation* (pp. 113–137). John Benjamins.
- Anderwald, L. (2008). The varieties of English spoken in the southeast of England: Morphology and syntax. In B. Kortmann & C. Upton (Eds.), *Varieties of English 1: The British Isles* (pp. 440–462). Mouton de Gruyter.
- Auer, P., Barden, B., & Grosskopf, B. (1998). Subjective and objective parameters determining ‘salience’ in long-term dialect accommodation. *Journal of Sociolinguistics*, 2(2), 163–187.
- Auer, P., & Hinskens, F. (2005). The role of interpersonal accommodation in a theory of language change. In P. Auer, F. Hinskens, & P. Kerswill (Eds.), *Dialect change: Convergence and divergence in European languages* (pp. 335–357). Cambridge University Press.
- Bergmann, C., Sprenger, S. A., & Schmid, M. S. (2015). The impact of language co-activation on L1 and L2 speech fluency. *Acta Psychologica*, 161, 25–35.
- Biber, D., Johansson, S., Leech, G., Conrad, S., & Finegan, E. (1999). *Longman Grammar of Spoken and Written English*. Longman.
- Biberauer, T. (2019). Factors 2 and 3: Towards a principled approach. *Catalan Journal of Linguistics*, 45–88.
- Blanchette, F., & Lukyanenko, C. (2019). Asymmetries in the Acceptability and Felicity of English Negative Dependencies: Where Negative Concord and Negative Polarity (Do Not) Overlap. *Frontiers in Psychology*, 10, 2486.
- Borer, H. (1984). *Parametric syntax: Case studies in Semitic and Romance languages*. Foris.
- Borrigan, M., De Bruin, A., Havas, V., De Diego-Balaguer, R., Vulchanova, M. D., Vulchanov, V., & Duñabeitia, J. A. (2021). Differences in word learning in children: Bilingualism or linguistic experience? *Applied Psycholinguistics*, 42(2), 345–366.
- Britain, D. (2002). Diffusion, levelling, simplification and reallocation in past tense BE in the English Fens. *Journal of Sociolinguistics*, 6, 16–43.
- Brown, R. (1973). *A first language: The early stages*. Harvard University Press.
- Buchstaller, I., Corrigan, K., Holmberg, A., Honeybone, P., & Maguire, W. (2013). T-to-R and the Northern Subject Rule: Questionnaire-based spatial, social and structural linguistics. *English Language and Linguistics*, 17(1), 85–128.

- Byland, E. (2009). Maturation constraints and first language attrition. *Language Learning*, 59(3), 687–716.
- Cabrera-Puche, M. (2008). *Null subject patterns in language contact: The case of Dominican Spanish* [PhD Thesis]. Rutgers University.
- Carroll, S. (2001). *Input and Evidence: The Raw Material of Second Language Acquisition*. John Benjamins Publishing Company.
- Cazzoli-Goeta, M., Guijarro-Fuentes, P., & Young-Scholten, M. (2010). Beyond attrition: What we need to consider in the study of new Spanish varieties. In P. Guijarro-Fuentes & L. Domínguez (Eds.), *New directions in language acquisition: Romance languages in the generative perspective* (pp. 453–478). Cambridge Scholars.
- Chambers, J. (1995). *Sociolinguistic theory: Linguistic variation and its social significance*. Blackwell.
- Chambers, J. K. (1992). Dialect Acquisition. *Language*, 68(4), 673–705.
- Chambers, J., & Trudgill, P. (2004). *Dialectology*. Cambridge University Press.
- Chatten, A., Baxter, K., Mas, E., Peña, J., Tabachnick, G., Duncan, D., & MacKenzie, L. (2022). “I’ve always spoke like this, you see”: Preterite-for-participle leveling in American and British Englishes. *American Speech*, 1–49.
- Cheshire, J. (1982). *Variation in an English dialect: A sociolinguistic study*. Cambridge University Press.
- Cheshire, J., Edwards, V., & Whittle, P. (1993). Non-standard English and dialect levelling. In J. Milroy & L. Milroy (Eds.), *Real English: The Grammar of English Dialects in the British Isles* (pp. 53–94). Longman.
- Cheshire, J., & Fox, S. (2009). Was/were variation: A perspective from London. *Language Variation and Change*, 21(1), 1–38.
- Chomsky, N. (1995). *The Minimalist Program*. MIT Press.
- Chomsky, N. (2000). *New Horizons in the Study of Language and Mind*. Cambridge University Press.
- Chomsky, N. (2001). Derivation by phase. In M. Kenstowicz (Ed.), *Ken Hale: A Life in Language* (pp. 1–51). MIT Press.
- Christensen, R. H. B. (2022). ‘ordinal’ package for R [R]. <https://cran.r-project.org/web/packages/ordinal/index.html>
- Conn, J. & Horesh, U. (2002). Assessing the acquisition of dialect variables by migrant adults in Philadelphia: A case study. *University of Pennsylvania Working Papers in Linguistics* 8(3), 47-57.
- Corrigan, K. (1997). *The syntax of South Armagh English in its socio-historical perspective*. University College Dublin.
- Corrigan, K. (2010). *Irish English vol. 1—Northern Ireland*. Edinburgh University Press.

- Coulson, S., King, J. W., & Kutas, M. (1998). Expect the Unexpected: Event-related Brain Response to Morphosyntactic Violations. *Language and Cognitive Processes*, 13(1), 21–58.
- Coupland, N., & Bishop, H. (2007). Ideologised values for British accents. *Journal of Sociolinguistics*, 11(1), 74–93.
- Cuza, A. (2010). On the L1 attrition of the present tense in Spanish. *Hispania*, 93(2), 256–272.
- Dayal, V. (1996). *Locality in wh-quantification: Questions and relative clauses in Hindi*. Kluwer.
- De Leeuw, E., Mennen, I., & Scobbie, J. M. (2013). Dynamic systems, maturational constraints and L1 phonetic attrition. *International Journal of Bilingualism*, 17(6), 683–700.
- de Leeuw, E., Schmid, M., & Mennen, I. (2010). The effects of contact on native language pronunciation in an L2 migrant setting. *Bilingualism: Language and Cognition*, 13(1), 33–40.
- Domínguez, L. (2013). *Understanding interfaces: Second language acquisition and first language attrition of Spanish subject realisation and word order variation*. John Benjamins.
- Domínguez, L., & Hicks, G. (2016). Synchronic change in a multidialectal community: Evidence from Spanish null and postverbal subjects. In A. Cuza, L. Czerwionka, & D. Olson (Eds.), *Inquiries in Hispanic linguistics: From theory to empirical evidence*. John Benjamins.
- Dube, S., Kung, C., Peter, V., Brock, J., & Demuth, K. (2016). Effects of Type of Agreement Violation and Utterance Position on the Auditory Processing of Subject-Verb Agreement: An ERP Study. *Frontiers in Psychology*, 7.
- Dussias, P. E. (2004). Parsing a first language like a second: The erosion of L1 parsing strategies in Spanish-English Bilinguals. *International Journal of Bilingualism*, 8(3), 355–371.
- Edwards, V. (1993). The grammar of Southern British English. In J. Milroy & L. Milroy (Eds.), *Real English: The Grammar of English Dialects in the British Isles* (pp. 214–238). Longman.
- Evans, B. (2004). The role of social network in the acquisition of local dialect norms by Appalachian migrants in Ypsilanti, Michigan. *Language Variation and Change* 16, 153–167.
- Fabricus, A. (2018). Social change, linguistic change and sociolinguistic change in Received Pronunciation. In N. Braber & S. Jansen (Eds.), *Sociolinguistics in England* (pp. 35–66). Palgrave Macmillan.
- Filppula, M. (1999). *The grammar of Irish English: Language in Hibernian style*. Routledge.
- Finlay, C. (1994). Syntactic variation in Belfast English. *Belfast Working Papers in Language and Linguistics*, 12, 69–97.
- Giles, H. (1979). *Language: Contexts and consequences*. Open University Press.
- Gimson, A. C. (1962). *An introduction to the pronunciation of English*. Edward Arnold.
- Gleitman, L. R., Newport, E. L., & Gleitman, H. (1984). The current status of the motherese hypothesis. *Journal of Child Language*, 11(1), 43–79.
- Goldschneider, J., & DeKeyser, R. (2001). Explaining the ‘natural order of L2 morpheme acquisition’ in English: A meta-analysis of multiple determinants. *Language Learning*, 51(1), 1–50.
- Green, L. J. (2002). *African American English: A linguistic introduction*. Cambridge University Press.

- Griffiths, J., & de Vries, M. (2013). The syntactic integration of appositives: Evidence from fragments and ellipsis. *Linguistic Inquiry*, 44(2), 332–344.
- Gürel, A. (2002). *Linguistic characteristics of second language acquisition and first language attrition: Overt versus null pronouns*. [PhD Thesis]. McGill University.
- Gürel, A. (2007). (Psycho)linguistic determinants of L1 attrition. In B. Köpke, M. Schmid, M. Keijzer, & S. Dostert (Eds.), *Language attrition: Theoretical perspectives* (pp. 99–119). John Benjamins.
- Gürel, A., & Yılmaz, G. (2011). Restructuring in the L1 Turkish grammar: Effects of L2 English and L2 Dutch. *Language, Interaction and Acquisition*, 2(2), 221–250.
- Heim, I. (1994). Interrogative semantics and Karttunen's semantics for 'know'. In R. Buchalla & A. Mittwoch (Eds.), *IATL 1* (pp. 128–144). Hebrew University of Jerusalem.
- Henry, A. (1995). *Belfast English and Standard English: Dialect variation and parameter setting*. Oxford University Press.
- Henry, A. (2002). Variation and syntactic theory. In J. Chambers, P. Trudgill, & N. Schilling-Estes (Eds.), *The Handbook of Language Variation and Change* (pp. 267–282). Blackwell.
- Henry, A. (2005). Idiolectal variation and syntactic theory. In L. Cornips & K. Corrigan (Eds.), *Syntax and Variation: Reconciling the biological and the social* (pp. 109–122). John Benjamins.
- Henry, A. (2016). Acquiring language from variable input: Subject-verb agreement and negative concord in Belfast English. *Linguistic Variation*, 16(1), 131–150.
- Henry, A., & Cottell, S. (2007). A new approach to transitive expletives: Evidence from Belfast English. *English Language and Linguistics*, 11(2), 279–299.
- Hickey, R. (2007). *Irish English: History and present-day forms*. Cambridge University Press.
- Hicks, G., & Domínguez, L. (2020). A model for L1 grammatical attrition. *Second Language Research*, 36(2), 143–165.
- Hicks, G., Domínguez, L., Jamieson, E., & Schmid, M. (2024). L1 grammatical attrition in late Spanish-English bilinguals in the UK: Aspectual interpretations of present tense in Spanish. *The Language Learning Journal*, 52(2), 145–158.
- Hinskens, F., Auer, P., & Kerswill, P. (2005). The study of dialect convergence and divergence: Conceptual and methodological considerations. In *Dialect change: Convergence and divergence in European languages* (pp. 1–50). Cambridge University Press.
- Hopp, H., & Schmid, M. (2013). Perceived foreign accent in first language attrition and second language acquisition: The impact of age of acquisition and bilingualism. *Applied Psycholinguistics*, 34, 361–394.
- Hsieh, L., Leonard, L. B., & Swanson, L. (1999). Some differences between English plural noun inflections and third singular verb inflections in the input: The contributions of frequency, sentence position, and duration. *Journal of Child Language*, 26(3), 531–543.
- Hughes, A., Trudgill, P., & Watt, D. (2012). *English Accents and Dialects* (5th ed). Taylor & Francis.

- Iverson, M. (2012). *Advanced language attrition of Spanish in contact with Brazilian Portuguese* [PhD Thesis]. University of Iowa.
- Jamieson, E. (2020). Viewing dialect change through acceptability judgments: A case study in Shetland dialect. *Glossa*, 5(1)(19), 1–28.
- Jones, D. (1918). *An outline of English phonetics*. Teubner.
- Kaan, E., Harris, A., Gibson, E., & Holcomb, P. (2000). The P600 as an index of syntactic integration difficulty. *Language and Cognitive Processes*, 15(2), 159–201.
- Keijzer, M. (2007). *Last in first out? An investigation of the regression hypothesis in Dutch emigrants in Anglophone Canada* [PhD Thesis]. Vrije Universiteit Amsterdam.
- Kerswill, P. (1994). *Dialects converging: Rural speech in urban Norway*. Clarendon.
- Kerswill, P., & Williams, A. (2002a). ‘Salience’ as an explanatory factor in language change: Evidence from dialect levelling in urban England. In M. Jones, E. Esch, & V. Herman (Eds.), *Language change: The interplay of internal, external and extra-linguistic factors* (pp. 81–110). Mouton de Gruyter.
- Kerswill, P., & Williams, A. (2002b). Dialect recognition and speech community focusing in new and old town in England: The effects of dialect levelling, demography and social networks. In D. Long & D. Preston (Eds.), *Handbook of Perceptual Dialectology Volume 2* (pp. 173–204). John Benjamins.
- Klima, E., & Bellugi, U. (1966). Syntactic regulation in the speech of children. In J. Lyons & R. Wales (Eds.), *Psycholinguistic papers: The proceedings of the 1965 Edinburgh conference* (pp. 183–203). Edinburgh University Press.
- Kolbe-Hanna, D. (2021). Variation in the complementiser choice between if and whether. In A. Čermáková & M. Malá (Eds.), *Variation in Time and Space: Observing the world through corpora* (pp. 203–230). de Gruyter.
- Köpke, B. (2007). Language attrition at the crossroads of brain, mind and society. In B. Köpke, M. Schmid, M. Keijzer, & S. Dostert (Eds.), *Language attrition: Theoretical perspectives* (pp. 9–37). John Benjamins.
- Köpke, B., & Schmid, M. S. (2004). Language attrition: The next phase. In M. S. Schmid, B. Köpke, M. Keijzer, & L. Weilemar (Eds.), *Studies in Bilingualism* (Vol. 28, pp. 1–43). John Benjamins Publishing Company.
- Kortmann, B., & Szmercsanyi, B. (2004). Global synopsis: Morphological and syntactic variation in English. In B. Kortmann, E. W. Schneider, C. Upton, K. Burridge, & R. Mesthrie (Eds.), *A handbook of varieties of English vol. 2* (pp. 1142–1202). Mouton de Gruyter.
- Labov, W. (1972a). Some principles of linguistic methodology. *Language in Society*, 1(1), 97–120.
- Labov, W. (1972b). *Sociolinguistic Patterns*. University of Philadelphia Press.
- Labov, W. (2001). *Principles of Linguistic Change* (Vols 2-Social Factors). Blackwell.

- Lahiri, U. (2002). *Questions and answers in concealed questions*. Oxford University Press.
- Lardiere, D. (2008). Feature-assembly in Second Language Acquisition. In J. Liceras, H. Zobl, & H. Goodluck (Eds.), *The role of formal features in second language acquisition* (pp. 106–140). Lawrence Erlbaum Associates.
- Lardiere, D. (2009). Some thoughts on the contrastive analysis of features in second language acquisition. *Second Language Research*, 25(2), 173–227.
- Legendre, G., Culbertson, J., Zaroukian, E., Hsin, L., Barrière, I., & Nazzi, T. (2014). Is children's comprehension of subject–verb agreement universally late? Comparative evidence from French, English, and Spanish. *Lingua*, 144, 21–39.
- Lidz, J., & Gagliardi, A. (2015). How nature meets nurture: Universal Grammar and statistical learning. *Annual Review of Linguistics*, 1, 333–353.
- Lindsey, G. (2019). *English After RP: Standard British Pronunciation Today*. Springer International Publishing.
- Lucek, S., & Garnett, V. (2019). Perceptions of linguistic identity among Irish English speakers. In R. Hickey & C. Amador-Moreno (Eds.), *Irish identities: Sociolinguistic perspectives* (pp. 145–178). de Gruyter.
- McCafferty, K. (2003). The Northern Subject Rule in Ulster: How Scots, how English? *Language Variation and Change*, 15, 105–139.
- McCloskey, J. (2006). Questions and questioning in a local English. In R. Zanuttini, H. Campos, E. Herburger, & P. Portner, *Crosslinguistic Research in Syntax and Semantics: Negation, Tense and Clausal Architecture* (pp. 87–126). Georgetown University Press.
- Milroy, J., & Milroy, L. (1985). Linguistic change, social network and speaker innovation. *Journal of Linguistics*, 21, 339–384.
- Milroy, L. (1980). *Language and social networks*. University Park Press.
- Milroy, L. (1984). Comprehension and context: Successful communication and communication breakdown. In P. Trudgill (Ed.), *Applied Sociolinguistics* (pp. 7–32). Academic Press.
- Montgomery, C. (2012). The effect of proximity in perceptual dialectology. *Journal of Sociolinguistics*, 16(5), 638–668.
- Montgomery, M. (2006). The morphology and syntax of Ulster Scots. *English World-Wide*, 27(3), 295–329.
- Mufwene, S. (1991). Pidgins, creoles, typology and markedness. In F. X. Byrne & T. Huebner (Eds.), *Development and structures of creole languages: Essays in honor of Derek Bickerton* (pp. 123–143). John Benjamins.
- Mufwene, S. (2001). *The ecology of language evolution*. Cambridge University Press.
- Murray, J. (1873). *The Dialect of the Southern Counties of Scotland*. Philological Society.

- Nevalainen, T. (2006). Vernacular universals? The case of plural was in early modern English. In T. Nevalainen, J. Klemola, & M. Laitenen (Eds.), *Types of variation: Diachronic, dialectal and typological* (pp. 351–369). John Benjamins.
- Nevins, A., & Parrott, J. K. (2010). Variable rules meet Impoverishment theory: Patterns of agreement levelling in English varieties. *Lingua*, 120, 1135–1159.
- Nycz, J. (2016). Awareness and acquisition of new dialect features. In A. Babel (ed.) *Awareness and control in sociolinguistic research*. (pp. 62-79). Cambridge University Press.
- Orfitelli, R., & Polinsky, M. (2017). When performance masquerades as comprehension: Grammaticality judgments in experiments with non-native speakers. In M. Kopotev, O. Lyashevskaya, & A. Mustajoki (Eds.), *Quantitative approaches to the Russian language*. Routledge.
- Osterhout, L., & Mobley, L. A. (1995). Event-Related Brain Potentials Elicited by Failure to Agree. *Journal of Memory and Language*, 34(6), 739–773. <https://doi.org/10.1006/jmla.1995.1033>
- Otheguy, R., & Zentella, A. C. (2012). *Spanish in New York: Language contact, dialectal leveling, and structural continuity*. Oxford University Press.
- Pietsch, L. (2005). “Some do and some doesn’t”: Verbal concord variation in the north of the British Isles. In B. Kortmann, T. Herrmann, L. Pietsch, & S. Wagner (Eds.), *A Comparative Grammar of British English Dialects* (pp. 125–210). Mouton de Gruyter.
- Poplack, S. (1980). Sometimes I’ll start a sentence in Spanish Y TERMINO EN ESPAÑOL: toward a typology of code-switching. *Linguistics*, 18, 581-618.
- Potts, C. (2005). *The Logic of Conventional Implicatures*. Oxford University Press.
- Pozzan, L., & Quirk, E. (2014). Second language acquisition of English questions: An elicited production study. *Applied Psycholinguistics*, 35(6), 1055–1086.
- Putnam, M., & Sánchez, L. (2013). What’s so incomplete about incomplete acquisition? A prolegomenon to modelling heritage language grammars. *Linguistic Approaches to Bilingualism*, 3(4), 478–508.
- Rothman, J., & Slabakova, R. (2018). The generative approach to SLA and its place in modern second language studies. *Studies in Second Language Acquisition*, 40(2), 417–442.
- Schifano, N. (2024). UK Ecuadorian Spanish: Mood and V-movement under contact. Presentation at *Romance grammars, contact and context*. University of Birmingham, January 2024.
- Siegel, J. (2010). *Second dialect acquisition*. Cambridge University Press.
- Schmid, M., & Dusseldorp, E. (2010). Quantitative analyses in a multivariate study of language attrition: The impact of extralinguistic factors. *Second Language Research*, 26(1), 125–160.
- Schmid, M., & Köpke, B. (2017). The relevance of first language attrition to theories of bilingual development. *Linguistic Approaches to Bilingualism*, 7(6), 637–667.
- Schmid, M. S. (2011). *Language Attrition*. Cambridge University Press.

- Schmitt, E. (2019). Morphological Attrition. In M. S. Schmid & B. Köpke (Eds.), *The Oxford Handbook of Language Attrition* (pp. 227–240). Oxford University Press.
- Schwartz, B. D., & Sprouse, R. A. (1994). Word order and nominative case in non-native language acquisition: A longitudinal study of (L1 Turkish) German interlanguage. In T. Hoekstra & B. D. Schwartz (Eds.), *Language Acquisition Studies in Generative Grammar* (pp. 317–368). John Benjamins.
- Schwartz, B. D., & Sprouse, R. A. (1996). L2 cognitive states and the Full Transfer/Full Access model. *Second Language Research*, 12(1), 40–72.
- Seliger, H. W., & Vago, R. M. (Eds.). (1991). *First Language Attrition*. Cambridge University Press.
- Sharma, D., Levon, E., & Ye, Y. (2022). 50 years of British accent bias: Stability and lifespan change in attitudes to accents. *English World-Wide*, 43(2), 135–166.
- Slabakova, R. (2016). *Second Language Acquisition*. Oxford University Press.
- Smith, J., & Tagliamonte, S. (1998). ‘We were all thegither... I think we was all thegither’: Was regularization in Buckie English. *World Englishes*, 17, 105–126.
- Spector, B., & Egré, P. (2015). A uniform semantics for embedded interrogatives: An answer, not necessarily the answer. *Synthese*, 192(6), 1729–1784.
- Stringer, D. (2015). Embedded wh-questions in L2 English in India: Inversion as a main clause phenomenon. *Studies in Second Language Acquisition*, 37(1), 101–133.
- Tagliamonte, S. (1998). Was/were variation across the generations: View from the city of York. *Language Variation and Change*, 10(2), 153–191.
- Tortora, C., & den Dikken, M. (2010). Subject agreement variation: Support for the configurational approach. *Lingua*, 120, 1089–1108.
- Trudgill, P. (1986). *Dialects in Contact*. Blackwell.
- Trudgill, P. (1999). Standard English: What it isn’t. In T. Bex & R. J. Watts (Eds.), *Standard English: The widening debate* (pp. 115–128). Routledge.
- Trudgill, P. (2008). Colonial dialect contact in the history of European languages: On the irrelevance of identity to new-dialect formation. *Language in Society*, 37(2), 241–254.
- Truscott, J., & Sharwood Smith, M. (2017). Representation, processing and code-switching. *Bilingualism: Language and Cognition*, 20(5), 903–916.
- Tsimpli, I., Sorace, A., Heycock, C., & Fillaci, F. (2004). First language attrition and syntactic subjects: A study of Greek and Italian near-native speakers of English. *International Journal of Bilingualism*, 8(3), 257–277.
- Ulbrich, C., & Ordin, M. (2014). Can L2-English influence L1-German? The case of post-vocalic /r/. *Journal of Phonetics*, 45, 26–42.
- Wells, J. C. (1982). *Accents of English*. Cambridge University Press.
- White, L. (1989). Linguistic universals, markedness and learnability: Comparing two different approaches. *Second Language Research*, 5(2), 127–140.

- White, L. (2003). *Second Language Acquisition and Universal Grammar*. Cambridge University Press.
- Woods, R. (2016). *The syntax of speech acts: Embedding illocutionary force* [PhD Thesis]. University of York.
- Woods, R. (2020). A different perspective on embedded verb second: Unifying embedded root phenomena. In R. Woods & S. Wolfe (Eds.), *Rethinking Verb Second* (pp. 297–322). Oxford University Press.
- Yılmaz, G., & Schmid, M. S. (2012). L1 accessibility among Turkish-Dutch bilinguals. *The Mental Lexicon*, 7(3), 249–274.

Appendix A – Principal Components Analysis

In the Principal Components Analysis (PCA) we began with 20 measures for each participant, which were responses to the sociolinguistic questionnaire. Five of these measures (the questions from Milroy's (1980) social networks questionnaire) had already been condensed into one measure of 'network density', and so were removed from the data, leaving 15 measures per participant. The measures are described in Table 1.

Table 1: Measures entered into the Principal Components Analysis

age	Participant age at time of data collection	affil_change	Has the participant's feelings of affiliation to England or Northern Ireland changed recently
move_age	Participant's age of arrival in Belfast	partner_from	Is the participant's partner from England, Northern Ireland or elsewhere
edu	Participant's education level	famL2	How much time does the participant spend with Northern Irish family
visit	How often participant visits England	friendsL2	How much time does the participant spend with Northern Irish friends
visit_length	For how long does the participant stay in England when visiting	workL2	How much time does the participant spend with Northern Irish colleagues
homesick	How homesick for England does the participant feel	network_density	A combined measure of how tight-knit and how Northern Irish a participant's social network is
moveback	Does the participant plan to move back to England		
affil_culture	Does the participant feel more affiliated with English or Northern Irish culture		

We ran a Bartlett test to ensure the data was suitable for PCA ($t = 155.05$, $df = 105$, $p = .0011$).

We used `prcomp()` to run the PCA in R. A scree plot was plotted with `fviz_eig` from the `factoextra` package, shown in Figure 1.

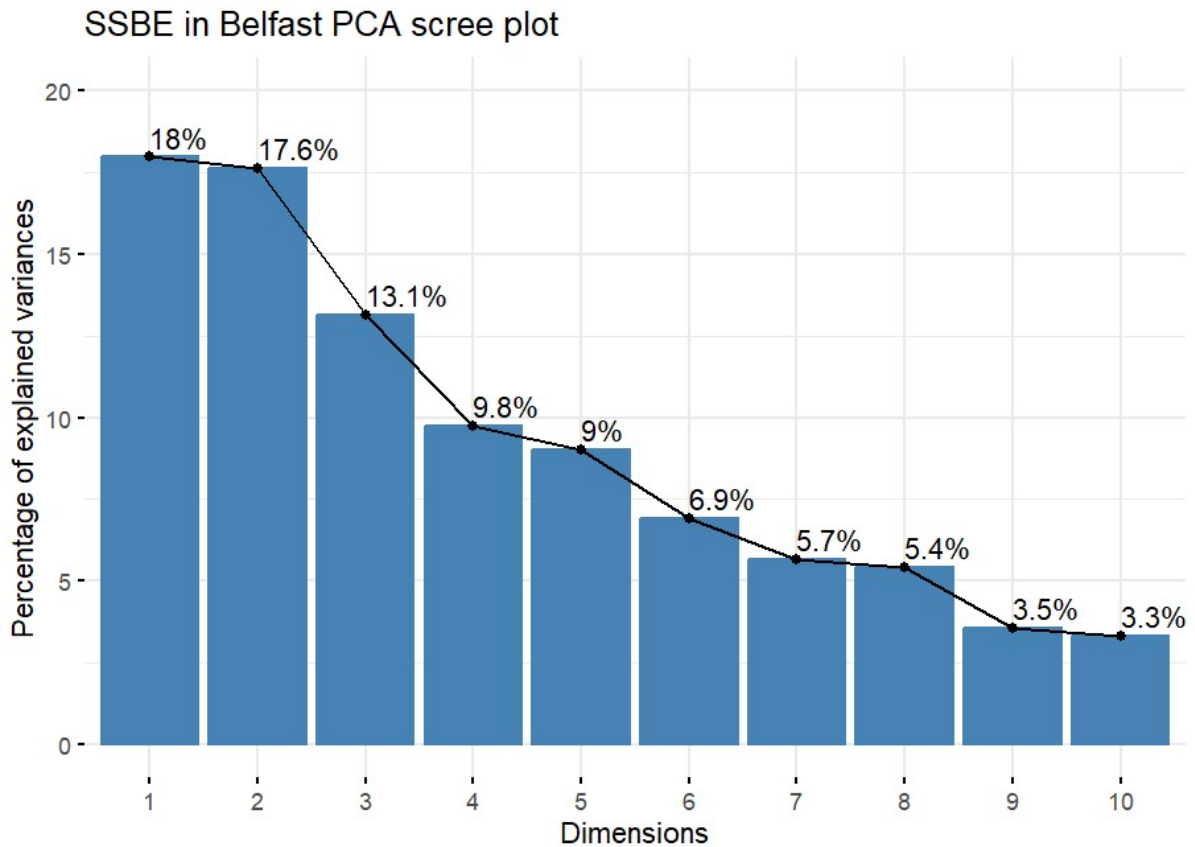


Figure 1: Scree plot of variance explained by the components in the PCA

We opted to focus on components which explained $> 10\%$ of the variance (i.e. components 1, 2 and 3). These also corresponded to the components for which eigenvalues were > 2 (eigenvalues of 2.7, 2.6 and 2 respectively). Figure 2 shows the most important measures contributing to each of the three components. The loadings for these three factors are shown in Table 2.

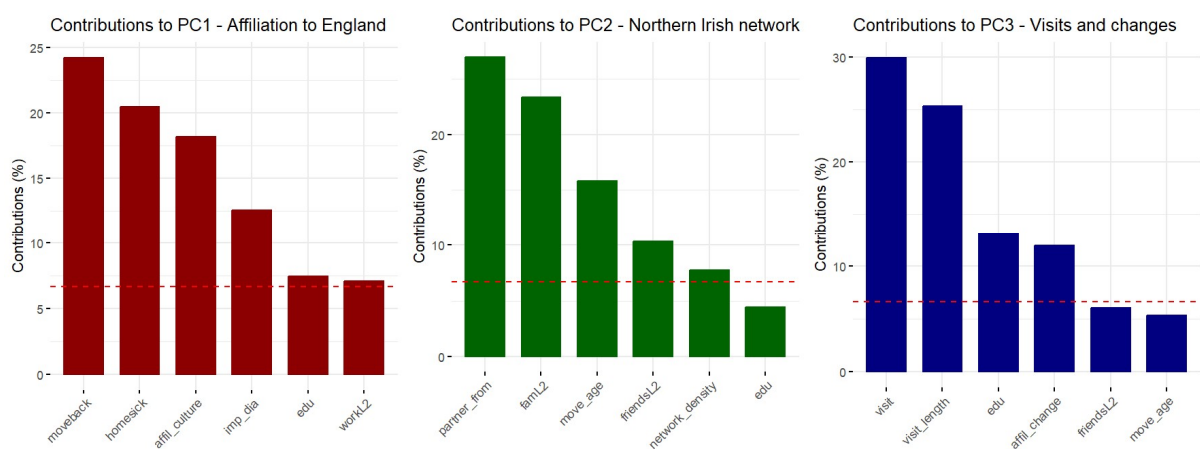


Figure 2: Variables contributing to the three principal components identified in our data

	PC1	PC2	PC3
age	-0.24	0.17	0.05
move_age	-0.06	0.40	0.23
edu	-0.27	0.21	-0.36
imp_dia	0.35	0.14	-0.12
visit	0.00	-0.15	-0.55
visit_length	0.12	-0.11	0.50
homesick	0.45	0.01	0.00
moveback	0.49	0.06	0.08
affil_culture	-0.43	-0.16	0.05
affil_change	-0.14	-0.04	0.35
partner_from	0.01	-0.52	-0.04
famL2	0.04	-0.48	-0.05
friendsL2	0.01	-0.32	0.25
workL2	-0.27	0.04	0.23
network_density	-0.05	-0.28	-0.02

Table 2: Loadings for the major components in the PCA

PC1 we label as ‘**Affiliation to England**’: the most important measures contributing to this component were desire to move back to England, homesickness for England, feeling more at home in English (rather than Northern Irish) culture, and the importance of maintaining their local (English) dialect. Other measures contributing to this component were the amount of time spent with Northern Irish people in the workplace, and the participant’s education level.

PC2 we label as ‘**Northern Irish network**’: the most important measures contributing to this component were whether the participant’s partner was from Northern Ireland, how much time they spent with friends and family from Northern Ireland, the density of their social network, and the age at which they moved to Belfast.

PC3 we label as ‘**Visits and change**’: the most important measures contributing to this component were how often and for how long participants visited England since moving to Belfast, as well as their education level and whether they had experienced any recent change in their feelings towards England vs Northern Ireland (as a result of e.g. Brexit or the Covid-19 pandemic, or more personal matters such as family relationships).